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Association of rapamycin treatment with the modulation of purine metabolism, reduced microglial inflammatory responses, improved mitochondrial energy metabolism, and alleviation of fatigue symptoms in ME/CFS subjects: pilot findings from phase-II observational study

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Abstract

Background and rationale In our ongoing phase II observational pilot trial, the compounded formulation of low-dose rapamycin significantly reduced fatigue-related clinical symptoms in ME/CFS subjects. Although the underlying molecular mechanism remains unclear, exploring metabolic pathways involving circulating blood-borne factors is warranted. Recent studies suggest that increased levels of purines may exacerbate oxidative stress in ME/CFS patients. It is not known if rapamycin modulates purine biosynthesis and improves disease symptoms.

Methods and results To address, we performed a comprehensive LCMS-based quantification of purine biosynthetic intermediates in plasma from responder cohort of ME/CFS participants, both at baseline (BSL) and after 90 days of rapamycin therapy (T3). Notably, differential regulation was observed in the enzymatic conversion of inosine monophosphate (IMP) to xanthosine-5-monophosphate (XMP) and hypoxanthine (HPX) between BSL and T3 samples. Flow cytometry assays on PBMCs confirmed that rapamycin reduces IMP dehydrogenase activity, thereby limiting the conversion of IMP to XMP. Further analyses, including mitochondrial oxidative stress assessments, Seahorse OCR following purine supplementation, and flow cytometry indicate that altered purine levels can impair mitochondrial energy metabolism, and may contribute to inflammatory processes in microglia.

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Conclusion Collectively, these findings highlight the therapeutic potential of rapamycin to enhance energy metabolism in patients with ME/CFS.

Major limitations There is no placebo group, and molecular results are somewhat biased to responders.

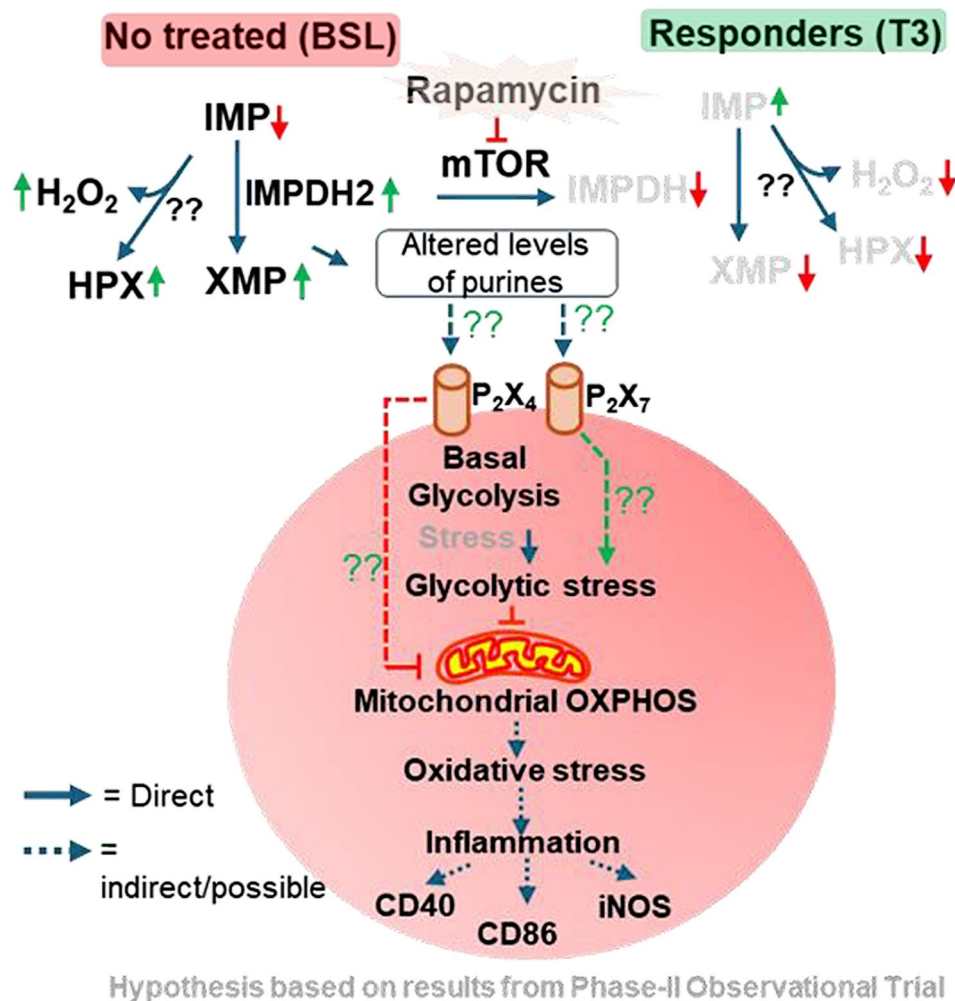
Trial registration CLINICALTRIALS.GOV, NCT06257420. Registered 11 December 2023, <https://clinicaltrials.gov/study/NCT06257420>.

Highlights

- A uniform compounded sirolimus (rapamycin) provides improved tolerability, fewer adverse effects, and sustained relief of ME/CFS symptoms over time.
- Rapamycin may modulate purine biosynthesis by inhibiting IMPDH2 activity, reducing IMP to XMP conversion in ME/CFS.
- Rapamycin may improve mitochondrial energy metabolism during stress in ME/CFS.
- Circulating purines are capable of activating purinergic receptors, which may exacerbate mitochondrial dysfunction and cellular stress in individuals with ME/CFS.

Keywords ME/CFS, PEM, mTOR, Rapamycin, Purines, Inflammation, Oxidative stress

Graphical Abstract



Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) [1] is a chronic, multisystem, and highly disabling disorder characterized by a broad and

heterogeneous spectrum of symptoms [2, 3]. Although disease onset varies considerably among patients, highlighting the heterogeneity of the condition [4], a common pattern involves the onset of illness following an

acute viral infection, including SARS-CoV-2, or the reactivation of ubiquitous beta-herpesviruses such as Epstein–Barr virus (EBV), cytomegalovirus (CMV), and human herpesvirus-6 (HHV-6) [3, 5, 6]. Clinically, ME/CFS patients frequently present with debilitating fatigue, myalgia [7], muscle weakness [8, 9], autonomic dysfunction [10] often manifesting as orthostatic intolerance [11], neurocognitive impairment [12], and post-exertional malaise (PEM) [13]. Increasing evidence suggests that these symptoms are associated with impairments in cellular energy metabolism [14], dysregulated innate and adaptive immune responses [15], increased oxidative stress signaling [16], loss of myelin integrity affecting skeletal muscle innervation [17], and neuroinflammation within the central nervous system [18, 19].

Post-exertional malaise (PEM) [13] is widely considered the hallmark symptom of ME/CFS and remains among the most poorly understood and difficult-to-treat aspects of the disease. PEM is characterized by a marked worsening of symptoms, including fatigue, pain, autonomic dysfunction, and cognitive impairment, following even minor physical or mental exertion. Symptom exacerbation is often delayed by 12–24 h after exertion and may persist for days or weeks. Pacing, an energy-conserving practice used by many ME/CFS patients, does not adequately prevent or relieve PEM [20]. Although the molecular mechanisms underlying PEM remain unclear, several of our recent studies suggest that a subset of patients with mTOR activation-mediated impaired autophagy may represent a central pathological feature of PEM [9, 21]. We further demonstrated that deficits in mitochondrial energy metabolism secondary to impaired autophagy may contribute to the profound muscle fatigue and neuromuscular dysfunction observed during PEM episodes [17]. Supporting this hypothesis, our recent phase I clinical observational study [21] demonstrated that low-dose generic rapamycin (6 mg/week), a selective mechanistic target of rapamycin (mTOR) inhibitor, was well-tolerated, safe, and significantly ameliorated PEM symptoms in ME/CFS subjects as measured by validated clinical endpoints while simultaneously improving biomarkers associated with autophagy function as measured by peripheral blood. These findings support the therapeutic potential of rapamycin in ME/CFS and implicate dysregulated mTOR-autophagy signaling in disease pathogenesis.

In addition to impaired autophagy, emerging evidence from our group suggests that abnormalities in purine biosynthesis may contribute to the pathophysiology of PEM. Purine biosynthesis [22, 23] is an essential metabolic process that generates nucleotide bases required for DNA and RNA synthesis [24, 25] and supplies critical intermediates involved in cellular bioenergetics, including ATP [26] and GTP [27]. Recent studies [28, 29] further indicate that excess extracellular purines may promote inflammation by activating purinergic P_2X [30] and P_2Y

[31] receptors expressed on myeloid [32] and lymphoid [33] immune cells. Consistent with this idea, a recent nuclear magnetic resonance (NMR)-based metabolomic study [34] identified altered purine metabolites in ME/CFS patients, suggesting dysregulated purine metabolism. More recently, we demonstrated that plasma from ME/CFS subjects increased the activity of the purine biosynthetic enzymes inosine monophosphate dehydrogenase 2 (IMPDH2) [35] and guanosine monophosphate synthetase (GMPS) [36] in human microglial cells under hypoxic conditions [34]. Furthermore, high-resolution metabolomic analyses [37] revealed significantly elevated levels of hypoxanthine (HPX), a purine intermediate that can promote reactive oxygen species generation by being converted to xanthine via xanthine oxidase [38]. Collectively, these findings suggest that dysregulated purine metabolism may contribute to inflammatory signaling and oxidative stress in ME/CFS and raise the possibility that rapamycin may modulate purine bioavailability and attenuate inflammation while restoring mitochondrial energy metabolism [35–39].

The present non-placebo-controlled observational study utilized a uniform, compounded formulation of rapamycin [40] designed to improve dosing consistency, absorption uniformity, and overall tolerability [41]. Participant retention in this decentralized “home-based” clinical trial was high, with an 83.33% study completion rate, allowing sustained longitudinal evaluation throughout the treatment period. Clinically, the study demonstrated consistent improvements across multiple validated outcome measures [42], including the Multidimensional Fatigue Inventory (MFI), the SF-36, the Specific Symptom Severity Scale (SSS), and additional NIH Common Data Elements instruments [43]. Importantly, the decentralized study infrastructure further enhanced protocol standardization by integrating direct-to-patient drug deliver, mobile phlebotomy for biospecimen collection, centralized safety monitoring through Quest Diagnostics, and pharmacokinetic/pharmacodynamic assessments to evaluate therapeutic drug exposure at both the individual and cohort levels [44].

While investigating the molecular mechanisms underlying these clinical responses, Seahorse extracellular flux analysis demonstrated that a 90-day rapamycin treatment significantly enhanced mitochondrial oxidative phosphorylation (OXPHOS) while reducing mitochondrial stress in peripheral blood mononuclear cells (PBMCs) derived from ME/CFS subjects. High-resolution LC-MS metabolomic analysis further revealed that low-dose rapamycin altered the bioavailability of several purine metabolites, including increasing inosine monophosphate (IMP) levels alongside reduced xanthosine 5-monophosphate (XMP) and hypoxanthine (HPX) concentrations. Flow cytometry analyses demonstrated

enhanced IMPDH2 activity in ME/CFS subjects at baseline, facilitating the conversion of IMP to XMP; however, rapamycin treatment inhibited IMPDH2, resulting in IMP accumulation and attenuation of the purine-associated inflammatory signaling. Finally, purine supplementation experiments, and siRNA-mediated gene silencing studies further demonstrated that altered circulating purines may directly promote inflammatory responses in myeloid cells, an effect that was substantially mitigated by rapamycin treatment.

Materials and methods

Clinical services and partners

All clinical services in this study were performed by AgelessRx (Ann Arbor, MI), a licensed pharmacy and clinical provider. Simmaron Research compensated AgelessRx for these services on behalf of all participants. Patients were recruited by their clinical Principal Investigators (PIs) and following informed consent and completion of baseline study procedures. Eligibility for treatment with compounded rapamycin was assessed by AgelessRx providers. Participants deemed eligible received a monthly supply of the study drug, which was shipped directly to their homes. Monthly shipments continued thereafter, contingent upon the participant's ability and willingness to complete required safety labs and other study milestones. AgelessRx provided prefilled "standing orders" (individual test request forms) for centralized safety laboratory testing (Quest Diagnostics) at predetermined study timepoints (baseline through T3 and beyond). These orders allowed participants to either present directly to a Quest facility or, more commonly, have safety labs collected concurrently with study samples via mobile phlebotomy services. The safety lab panel was taken every 30 days after baseline). Safety lab results for each participant at every timepoint were first reviewed by the ordering provider at AgelessRx and then securely transmitted to Simmaron study staff through a HIPAA-compliant, IRB-approved data transfer portal. Safety labs were also independently monitored by each participant's clinical PI. Clinical PIs and AgelessRx providers worked collaboratively to review safety data and communicated with the overall study PI when intervention was warranted, including consideration of repeat testing or temporary discontinuation of therapy. Notably, no participants in this study presented safety laboratory findings that precluded initiation of therapy, nor did any laboratory abnormalities during the study necessitate a hold or discontinuation of treatment. Additionally, AgelessRx and study PIs jointly evaluated potential contraindications to rapamycin therapy, including concomitant medications. One participant was excluded from the trial due to daily use of hydrocortisone and fludrocortisone. Overall, this integrated clinical oversight model ensured

that all required safety testing for both initial screening and ongoing monitoring was completed, supporting the safe administration of the study drug throughout the protocol.

Workflow of clinical studies

For this clinical trial, participants completed a 90-day drug study period in which prior to starting the drug, participants completed consent documents, baseline questionnaires, and a baseline blood draw. After completing these baseline tasks, the participant was screened to ensure that they would be able to participate in the trial, as described above. Once the participant started the drug, the 90-day initial trial period began, in which the participants took their dose of rapamycin weekly and completed questionnaires and blood draws every 30 days for the trial period (Supplementary Fig. 1A). A supplementary table (Supplementary Fig. 1B) lists the questionnaires and instruments used at each timepoint in this study (BSL-T3), all of which are recommended for use as objective endpoints in clinical studies by the NIH Common Data Elements disease specific questionnaire library. The medical history form and the "Questionnaire for ME/CFS Participants (QMEP)" collected at the baseline served to provide previous medical history data pertinent to determine subsets for post hoc analyses (i.e. viral or non-viral onset, date of diagnosis, comorbid conditions etc.). The total and individual scored components of the SF-36, the MFI, and the BAS served as the study's primary endpoints. The FUNCAP, FACIT-F, FACT-G, and the SSS served as this study's secondary endpoints. See *Statistical Methods* for detailed methodology. De-identified, responses and related study data (safety lab results etc.) were maintained in the Simmaron IRB approved REDCAP server housed at Cornell University that was only accessible as read only to study staff executing the data analysis.

Recruitment of human subjects and rapamycin treatment

In our 2nd phase of rapamycin trial, all de novo ME/CFS subjects were enrolled according to the 2003 Canadian Consensus and IOM criteria, as previously discussed. All participants had to be between the age of 18–85 years old and reside in the United States to be enrolled in the trial. This group of 108 consented participants, currently involved in the second phase of the rapamycin trial, represents a new cohort receiving a different formulation and dosage schedule of rapamycin treatment.

Written informed consent was obtained as described earlier [21]. Criteria to categorize participants into three categories – responders, partial responders, and non-responders – is the same as in Phase I of the clinical trial. Responder status was determined using SF-36, MFI, and

BAS questionnaires that were taken at BSL through T3 (Table 1).

This current study is focused on the metabolic changes and improvement in clinical symptoms in these post-infectious subjects. Based on the self-reported medical history and illnesses survey, which was taken by participants prior to starting the trial, of the participants who began taking the trial regimen of rapamycin 51.32% ($N=39$) of participants experienced a known viral onset of ME/CFS and 48.68% ($N=37$) did not experience a known viral onset.

Machine learning model

As explained previously [21], a machine learning model was used to help assess the generalized cohort after the model was applied to non-completer forms. In this stage of the trial, the same exploratory machine learning model was used to determine responder status based on responses to health questionnaires and to predict responder status based on initial questionnaire responses.

Questionnaires

As displayed in Supplementary Fig. 1B, participants completed 9 repeated questionnaires at each timepoint

Table 1 Responder status criteria determined by primary endpoint measures. The responder status of participants was determined by their questionnaire responses to the Primary Endpoint Measures including the SF-36, MFI, and BAS. This was also how overall responder status was determined

Primary Endpoint Measure	Responder	Partial Responder	Non-Responder
SF-36	Meets 2 or more of the following: · $\geq +3$ VT sub-scale · $\geq +3$ SF sub-scale · $\geq +3$ RP sub-scale · $\geq +2$ PCS sub-scale	Meets 1 of the responder criteria	Meets none of the responder or partial responder criteria
MFI	Meets either of the following: · ≤ -2 in 3 or more of the 5 fatigue domains · ≤ -2 in 2 fatigue domains and MFI aggregate ≤ -10	Meets either of the following: · ≤ -2 in 2 or more of the 5 fatigue domains · ≤ -2 in 1 fatigue domains and $-10 < \text{MFI aggregate} < 0$	Meets none of the responder or partial responder criteria
BAS	$\geq +10$ in BAS	-----	≤ 0 in BAS
Overall Responder Status	≥ 2 Primary Endpoint Measures (SF-36, MFI, or BAS) classified as Responder	1 Primary Endpoint Measure (SF-36, MFI, or BAS) classified as Partial Responder , but not classified as responder overall	Meets none of the responder or partial responder criteria

(BSL-T3). BAS, SSS, MFI, SF-36, Sleep Questionnaire, and 24-Hour Health Survey explained previously [21]. Surveys that were new to this phase of the trial included the DePaul Symptom Questionnaire’s for both COVID and PEM, the FACIT-F, Functional Capacity (FUNCAP55), Sleep questionnaire, and the Wearable Device form.

Wearable Device Questionnaire: Participants at their baseline completed a wearable device questionnaire, stating whether or not they wore a fitness tracking device (Garmin, Apple Watch, Oura Ring, Visible, etc.). Initially, they were asked if their device tracked any of the following criteria: VO2 Max, Activity Score, Daily Average Step Count, and Sleep. If the answer was yes, then the participant was prompted to fill out said data for the previous 30 days. Participants would continue to fill out this data at each timepoint BSL-T3. At T3, participants were asked to upload the wearable device’s health data, allowing our team to analyze heart rate variability (HRV) and sleep data at each timepoint.

FACIT- F (Functional Assessment of Chronic Illness Therapy- Fatigue): Physical well-being, Social well-being, Emotional well-being, Functional well-being, and Fatigue are measured.

FACT-G (Functional Assessment of Cancer Therapy-General): Sections include Physical well-being, Social well-being, Emotional well-being, Functional well-being. These four sections are scored and then combined for the “General” score.

Safety lab and study sample collection

Once the baseline questionnaires were completed, four study kits (BSL-T3) and four safety kits were shipped to the participants’ provided address prior to scheduling the blood collection appointment. Once the kits arrived, the participants were instructed on how to schedule a mobile phlebotomy appointment via Travalab- or if Travalab did not cover their area of residence, a local mobile phlebotomist was scheduled to complete the blood draw. Each study kit included the following BD Vacutainer® tubes (Franklin, NJ, USA): one 2.5 mL PaxGene® blood RNA tube, one 2.7 mL sodium citrate (NaCit) tube, one 8.5 mL SST tube, and two 10.0 mL K2EDTA tubes. Each safety kit included the following BD Vacutainer® tubes (Franklin, NJ, USA): three 8.5 mL SST tubes and two 10.0 mL K2EDTA tubes. All SST tubes in both the study kit and safety kit were spun by the mobile phlebotomist at 1200 rcf for 10 min prior to shipping both kits to the Simmaron Research and Development Laboratory (University of Wisconsin-Milwaukee) overnight via FedEx. All safety lab samples were submitted to Quest Diagnostics within 24 h of collection for analysis. Safety labs processed at Quest Diagnostics included Hemoglobin A1C (#496), Insulin (#561), CBC (#6399, Includes Differential and

Platelets), Lipid Panel (#7600, Lipid Panel, Standard), Hs-CRP (#10124), Comprehensive Metabolic Panel (#10231), and for women under 50 years of age HCG (#84350 HCG, Total, Quantitative). AgelessRx® monitored the results. All study kit samples were processed under Good Laboratory Practice (GLP) conditions within 12–24 h of collection; processing was completed at the Simmaron Research and Development Laboratory (University of Wisconsin-Milwaukee).

Biobanking

From whole blood, plasma, serum, PaxGene® RNA, and NaCit-platelet depleted plasma were extracted, aliquoted, and immediately stored at -80 °C. To support longitudinal follow-up, a biobank library was established, in which each participant was assigned an individual storage box containing the cryopreserved samples. This system ensured that there was clear traceability of each sample. Since each participant functioned as their own control, this individualized library system allowed for efficient access to participant samples at each timepoint (BSL-T3).

Upon receipt of the participant blood samples, whole blood was taken from one K2EDTA tube was used for isolating PBMCs using the STEMCELL EasySep™ Direct Human PBMC Isolation Kit (catalog #19654; STEMCELL) protocol. Cell count and viability was determined using a Countess 3 Automated Cell Counter (ThermoFisher) after adding 1:1 (v/v) enriched cell suspension to trypan blue to a Countess Cell Counting Chamber Slide (catalog #10228; ThermoFisher). Cell count and viability were then saved on a hard disk. The enriched cell suspension was then added to a centrifuge tube, and equal parts of PBS was added to the tube. The sample tube was then spun in a swinging bucket centrifuge at 300 rcf at 20 °C for 10 min. After completion, the supernatant was aspirated and the pellet was re-suspended in CryoStor® CS10 (catalog #100–1061; STEMCELL) and aliquots were made with approximately 5 million live cells per 1mL CryoStor®. The aliquots were then placed into 4 °C for 10 min, then transferred into a Mr. Frosty and placed inside a -80 °C freezer for 24 h. After 24 h the samples were then placed into liquid nitrogen.

Seahorse assay

For PBMCs

For preparation of the Seahorse Mito Stress Test, samples were pulled out of liquid nitrogen and placed directly on ice. Once thawed, cell count and cell viability was checked using the same protocol as in PBMC isolation. Cells were then plated based on cell density with approximately 400,000 live cells per well. The 96-well Seahorse plate was then spun at 300 rcf for 5 min, afterwards cell adhesion was checked. Excess CryoStor® from the cell solution was aspirated out of the plate and Seahorse XF

Base Media was added to each well. Seahorse XF Cell Mito Stress assay was then performed using injections of 1 µM Oligomycin, 1.5 µM FCCP, and 0.5 µM Rotenone + Antimycin A.

For C20 Microglia

C20 Microglia were plated into the Seahorse XF-96 plate and allowed to grow for 24 h at 36 °C up to approximately 200,000 cells per well. After 24 h, the microglia were treated with serum-free DMEM supplemented with 10% serum samples (BSL, T3, and HC). Then the 96 well plate was placed into a hypoxia chamber (10% CO₂, 5% O₂, and 85% N₂) for 24 h. Before the assay, the existing media was washed and then replaced with assay-specific XF media. The Mito Stress assay media was XF base media with 10 mM glucose, 1 mM pyruvate, and 2 mM glutamine. The Glycolysis Stress assay media was XF base media with 200mM glutamine. Both the Mito Stress and Glycolysis Stress Tests were then run on the Seahorse XF-96 Analyzer. For Mito Stress, the assay was performed using injections of compounds as described under “PBMC Seahorse Assay”. The Glycolysis Stress Test assay was performed using injections of 10 mM Glucose, 1 µM Oligomycin, and 50 mM 2-DG (Deoxy-D-Glucose).

Si-RNA Transfection Studies of P₂X₄ and P₂X₇

C20 microglial cells were cultured on a 6-well plate. Once reached 80% confluency, cells were transfected with 5 picomoles P₂X₄ and P₂X₇ siRNA (ThermoFisher AM16708) via Lipofectamine™ 3000 transfection reagent (ThermoFisher L3000-001) as per the manufacturer's protocol. After 4 h of incubation, cells were treated with serum-free DMEM supplemented with 10% serum samples (HC, BSL and T3) and placed in a hypoxia chamber supplied with 85%N₂: 10% CO₂: 5% O₂ gas blend for an additional 24 h and subjected flow cytometry/FACS analysis.

FACS analysis

Flow cytometry analysis (FACS) was performed using the BD Accuri™ C6 Plus (BD Biosciences, San Jose, CA) flow cytometer instrument for cell sorting, data collection, and the FlowJo v10.10.0 software for data analysis. The detailed procedure was described elsewhere [17]. Myeloid cells-specific antibodies against CD11b (eBioscience FITC 11-0118-42), CD40 (eBioscience APC 17-0409-42), CD86 (eBioscience PE 12-0869-42) and CD206 (eBioscience APC 17-2069-42) were used. For FACS analysis, we used 1 × 10⁵ cells, centrifuge at 12000 rpm for 10 minutes. Cells were washed and resuspended in 100 µL of cold FACS buffer (and incubated with 2 µL of respective fluorophore-conjugated antibody for 30 minutes at rotator in the dark. After incubation, cell antibody complexes were centrifuged, washed, and the pellet

Table 2 The details of Internal controls and standards used in LCMS study

Compounds	Vendor	Catalog #	Application
8-Bromoadenosine 5'-monophosphate	MilliporeSigma	B3131-25MG	Internal control
5-(p-methylphenyl)-5-phenylhydantoin	SANTA CRUZ BIOTECHNOLOGY INC	sc-210,245	Internal control
Xanthosine Monophosphate	FISHER SCIENTIFIC	502,453,981	Standard
Hypoxanthine	Millipore Sigma	H9377-5G	Standard
Inosine Monophosphate	Millipore Sigma	I2879-1G	Standard

was resuspended in 200 µL of FACS and analyzed by flow cytometry.

LCMS method for the detection of purines in plasma samples

The following (Table 2) outlines the instrument configuration, mobile phase composition, column calibration procedures, and chemical components used for various internal controls. Subsequently, comprehensive details regarding sample preparation and analytical procedures are provided.

Pump configuration and setup

The pump setup for the LC system [45], including flow rate parameters and solvent routing. A C18 column (150 × 2 mm) operated at 0.3 mL/min is used. Proper pump configuration ensures stable pressure, consistent elution, and robust performance across all injections. The system is tuned to minimize carryover and maintain retention time stability.

Mobile phase composition

The gradient profile used for LC MS/MS separation. Mobile Phase A consists of 3 mM ammonium formate, while Mobile Phase B is methanol. The gradient program was developed and validated to ensure adequate separation of metabolites including Hypoxanthine, IMP, and XMP (Table 2). Smooth transitions between low and high

organic content help optimize retention, peak shape, and reproducibility.

Calibration curve and elimination of matrix effect

The preparation of an eight-point calibration curve spanning 1–1000 ppb. Calibration levels include: 1, 5, 10, 25, 50, 75, 250, 500, and 1000 ppb. A fixed internal standard at 100 ppb is spiked into each calibration level to ensure consistency and accurate quantification. A blank sample is also prepared to assess background signal and verify absence of contamination. The linearity of the calibration curve was measured by calculating the correlation coefficient from a liner curve between increasing concentration and area under the peak which was shown in Table 3. In order to eliminate the matrix effect, indirectly understand the physiological relevance of these metabolites, and validate the accuracy of the analysis, we performed spike-in-recovery measured by calculating % recovery after spiking the metabolite with the addition of 250 ppb metabolites externally. The formula used is $\% Recovery = \frac{(\text{spiked mean}) - (\text{unspiked mean})}{250}$. The validation confirms that both XMP and HPX can be detected accurately with minimum matrix effects, whereas IMP level is relatively low but falls within the acceptable range.

Plasma sample preparation protocol for LCMS analysis

1. Sample Preparation. Plasma samples were thawed on ice. For each experiment, 75 µL of plasma was transferred into a 1.5 mL Eppendorf tube. To each sample, 25 µL of the internal standard (8-bromo AMP, 4 ppm) was added. For spike-in experiments, an additional 25 µL of a 2 ppm working solution of each metabolite (three total metabolites) was added. Samples were vortexed briefly to ensure homogeneity. Details of internal control and standard metabolites were listed in Table 2.
2. Protein Precipitation. Protein precipitation was initiated by adding 25 µL of 0.1% formic acid to each tube, followed by vortex mixing for 5 seconds. Subsequently, 375 µL of ice-cold methanol was

Table 3 Detection statistics of internal controls and standards

Compounds	Precursor (m/z)	Product (m/z)	RT (min)	Corr, Coefficient	Linearity Range (ppp)	Spike Recovery (%) (spike 250 ppb)	Accuracy of the method
8-Bromoadenosine 5'-monophosphate	425.90	213.95	6.10	N/A	N/A	N/A	N/A
5-(p-methylphenyl)-5-phenylhydantoin	267.10	196.05	10.40	N/A	N/A	N/A	N/A
Xanthosine Monophosphate	364.90	97.10	2.50	0.99552	1-1000	157.2-185.4	reliable (> 100%)
Hypoxanthine	137.00	55.10	2.70	0.98793	1-1000	471-1576.4	reliable (> 100%)
Inosine Monophosphate	349.10	137.05	2.60	0.99147	1-1000	49.7-83.9	Low, but within reliable limit (50-85%)

added to each sample to complete precipitation. The mixture was vortexed for an additional 15 seconds.

3. Incubation. The samples were incubated at -20°C for 30 minutes to enhance protein aggregation.
4. Centrifugation. Following incubation, samples were centrifuged at $10,000 \times g$ for 10 minutes at 4°C . The resulting clear supernatants were carefully transferred to clean tubes without disturbing the precipitated protein pellet.
5. Filtration and Concentration. Supernatants were filtered through $0.2\ \mu\text{m}$ syringe filters into TurboVap-compatible tubes. The filtrates were concentrated under a nitrogen stream using TurboVap at a flow rate of $1.2\ \text{mL}/\text{min}$ and 25°C . Samples were evaporated until a thin film remained.
6. Reconstitution. Dried samples were reconstituted to a final volume of $200\ \mu\text{L}$ using LC-MS grade water, and 10 ppm 5-ME-8 internal standard was added (e.g., $20\ \mu\text{L}$ of 10 ppm stock with $180\ \mu\text{L}$ water). Samples were vortexed and transferred into autosampler vials with inserts.
7. LC-MS Analysis. Reconstituted samples were analyzed using LC-MS under optimized chromatographic and mass spectrometric settings. Analytes were quantified using calibration curves with internal standard correction. All samples were kept at 4°C in the autosampler prior to injection. Characterizations of peaks and their retention times were listed in Table 3.

IMPDH activity assay

To measure IMPDH activity, an IMPDH enzyme assay kit (Cat #: E-119) was obtained from Biomedical Research Service at the University of Buffalo. Around 10^5 PBMCs were lysed using $300\ \mu\text{L}$ of 1X sample lysis buffer, and protein amounts were determined with the Bradford method. The reaction solution stock was made by combining one part of 50x IMPDH Substrate (included in the kit) with fifty parts of IMPDH Assay Solution. For the control solution stock, one part of dH₂O was mixed with fifty parts of IMPDH Assay Solution (without substrate). Control and reaction wells were arranged in two rows on a 96-well plate. Twenty microliters of each sample lysate were added to each well, followed by $50\ \mu\text{L}$ of control or reaction solutions as appropriate. The plate was wrapped in aluminum foil and incubated at 37°C in a humidified environment for 60 min. Readings in both control and reaction wells were taken at OD₄₉₂ nm every three minutes for thirty minutes. The normalized OD ($\Delta\text{OD} = \text{OD reaction} - \text{OD control}$) was converted into $\text{ng mg}^{-1}\text{hour}^{-1}$ units using the formula provided with the kit. The detection range aligns with physiological levels reported in other studies.

Immunofluorescence (IF) analysis:

C20 microglial cells were plated in a chamber slide with complete DMEM media for 24 h to reach the desired ($\sim 70\%$) confluency. After 24 h the media was replaced with serum free DMEM supplemented with 10% plasma samples and incubated in a hypoxia chamber filled with a mixture of $85\%\text{N}_2 + 10\%\text{CO}_2 + 5\%\text{O}_2$ gases. After 24 h, the plate was taken out and cells were assayed IF analyses. Cells were fixed with chilled methanol overnight, washed with 1X PBS, blocked with 2% BSA containing blocking buffer for 30 min, incubated with primary antibodies (1:100–250) dilution at 37°C for 2 h, washed (3X) with wash buffer, incubated with either FITC- or TRITC-labeled anti-host (2°) secondary antibody for 1 h at 37°C , washed, stained nuclei with DAPI (1:10000 dilution), and imaged at Accuscope fluorescence microscope.

Power analysis

Power analysis was conducted to determine the n required to achieve sufficient power. The primary difference from baseline to T3 was derived from pilot Dunnett's data. To achieve 80% power with the most conservative measures (accounting for multiple comparisons, $\alpha = 0.0167$, half-effect, $d = 0.50$), the inflated sample size at baseline is 78 with 65 completers at T3 (83.33% retention), exceeding the planned completer count and thus providing at least the target power for the prespecified effect size.

Statistical analysis

Percentages and p-values from the chi-square test of independence were calculated for categorical variables. The mean, range, SEM, and p-values from Student's t-test were calculated for continuous variables. Summary data analysis was performed using Python version 3.12.8, pandas version 3.0.1, and SciPy version 1.17.1.

Changes in patient-reported outcomes measures were plotted using column graphs with individual points and SEM bars. A mixed effects model was fit using restricted maximum likelihood (REML), similar to a repeated-measures ANOVA, and the p-values were reported. Geisser-Greenhouse correction was used to correct for the assumption of sphericity in all variance analyses. Between-group comparisons were made using Dunnett's test for each time point versus the baseline. The two-stage step-up method of Benjamini, Krieger, and Yekutieli (BKY FDR) was used to correct for the false discovery rate ($\alpha = 0.05$) for all analyses. Statistical significance was determined based on the resulting FDR-adjusted p-values. QQ plots were plotted for each mixed-effect model to assess the assumption of normal residuals. Statistical significance was set at p less than 0.05. Inferential data analysis was performed using GraphPad Prism 11 version 11.0.1.

Results

Effect of rapamycin on the alleviation of clinical symptoms of fatigue in ME/CFS subjects: The design of Phase-II trial, execution, data acquisition, and analysis of clinical symptoms

Sex-specific dosing strategy and dose-ramping approach

The phase II rapamycin trial consisted of four major study milestones and observed patients for 90 days on compounded low-dose rapamycin: baseline (BSL), Timepoint 1 (T1), Timepoint 2 (T2), and Timepoint 3 (T3), with sex-specific dose-escalation strategies implemented for both male and female participants. For female participants, T1 was reached after 30 days of treatment and marked completion of a four-week dose-ramping schedule in which rapamycin dosing increased sequentially from 2.5 mg/week to 5 mg/week, then to 10 mg/week, ultimately reaching a maintenance dose of 15 mg/week through T3 (90 days) (Supplementary Fig. 1A). Male participants initiated treatment at 5 mg/week and increased dosing by 5 mg/week each week until reaching a maintenance dose of 20 mg/week through T3.

The rapamycin utilized in this study was a compounded formulation manufactured by Precision Compounding Pharmacy (Bellmore, NY, USA) and distributed by AgelessRx. It is reported that compounded versions of rapamycin are three times less bioavailable compared to generic formulation, thus a dose of 15–18 mg/week is equivalent to 5–6 mg/week of the generic formulation. It is also reported that compounded formulations of rapamycin have slightly fewer adverse events, particularly in the dose escalation phase. Dose selection and escalation schedules were informed by prior safety laboratory findings using the same compounded formulation in large independent clinical trials in healthy adults [41, 46], and in conjunction with recommendations from collaborating clinicians and pharmacologic validation performed with AgelessRx.

Recruitment of participants, timeline, and drop-outs

A total of 108 participants were consented and screened on a rolling basis between January of 2025 and March of 2026 (Supplementary Fig. 2). Following completion of baseline questionnaires and blood collection, 20 participants elected not to initiate therapy, while an additional 4 participants did not meet independent medical screening criteria for rapamycin treatment. Of the remaining 84 eligible participants, 78 initiated rapamycin treatment, whereas 6 participants had not yet begun treatment at the time of manuscript submission.

Among the 78 participants who initiated therapy, 3 patients discontinued treatment during the dose-ramping phase prior to T1 due to adverse events that included: non-specific muscle twitching, gastrointestinal symptoms

(diarrhea), and erythematous dermatitis. 75 participants completed treatment through T1. Between T1 and T2, 1 participant discontinued therapy due to tenosynovitis, while three additional participants initiated an ongoing drug holiday for personal reasons unrelated to study safety but prevented their continued participation in the trial. 1 participant remained actively enrolled between T1 and T2 at the time of analysis. 70 participants completed T2, with 1 additional participant beginning an ongoing drug holiday thereafter (unrelated to study safety). 4 participants remain on the study between T2 and T3 at the time of manuscript submission.

Overall, 65 participants completed the full BSL-to-T3 protocol. Analysis of the primary endpoints identified that among these completers, 26 participants were classified as responders, 31 as partial responders, and 5 as non-responders. An additional 11 participants remained actively enrolled at the time of manuscript submission. Unless otherwise indicated, all analyses were performed using the combined cohort of trial completers and active participants ($N=76$). Demographic and clinical characteristics of this cohort are summarized in Supplementary Fig. 3.

Pharmacokinetic analysis of compounded rapamycin

Pharmacokinetic analyses were performed in 25 ME/CFS participants to assess rapamycin bioavailability following weekly dosing (15 mg/week in females; 20 mg/week in males) (Supplementary Fig. 4). Blood sirolimus concentrations were measured on the day of dosing (Day 0) and on days 1, 2, 3, 5, and 6 following dose administration.

Among participants assessed on the day of dosing or peak ($N=21$), the mean plasma sirolimus concentration was 9.643 ± 2.855 ng/mL. Mean concentrations subsequently declined to 2.756 ± 0.627 ng/mL on Day 1 ($N=9$), 1.45 ± 0.233 ng/mL on Day 2 ($N=10$), 1.0 ± 0 ng/mL on Day 3 ($N=3$), 1.0 ± 0 ng/mL on Day 5 ($N=1$), and 1.1 ± 0 ng/mL on Day 6 ($N=1$). Overall, these findings recapitulate previously published pharmacokinetic data from large populations and FDA-approved studies of rapamycin. Plasma sirolimus concentrations measured at the approximate peak and trough were comparable to those previously reported for both generic and compounded formulations in healthy adult populations.

Safety lab analysis

No statistically significant differences were observed in any safety laboratory parameters between baseline and T3 (Supplementary Fig. 5). We also did not observe any significant increase in reported infections throughout the study period. All reported adverse events (Table 4) were non-serious and resolved shortly after treatment discontinuation.

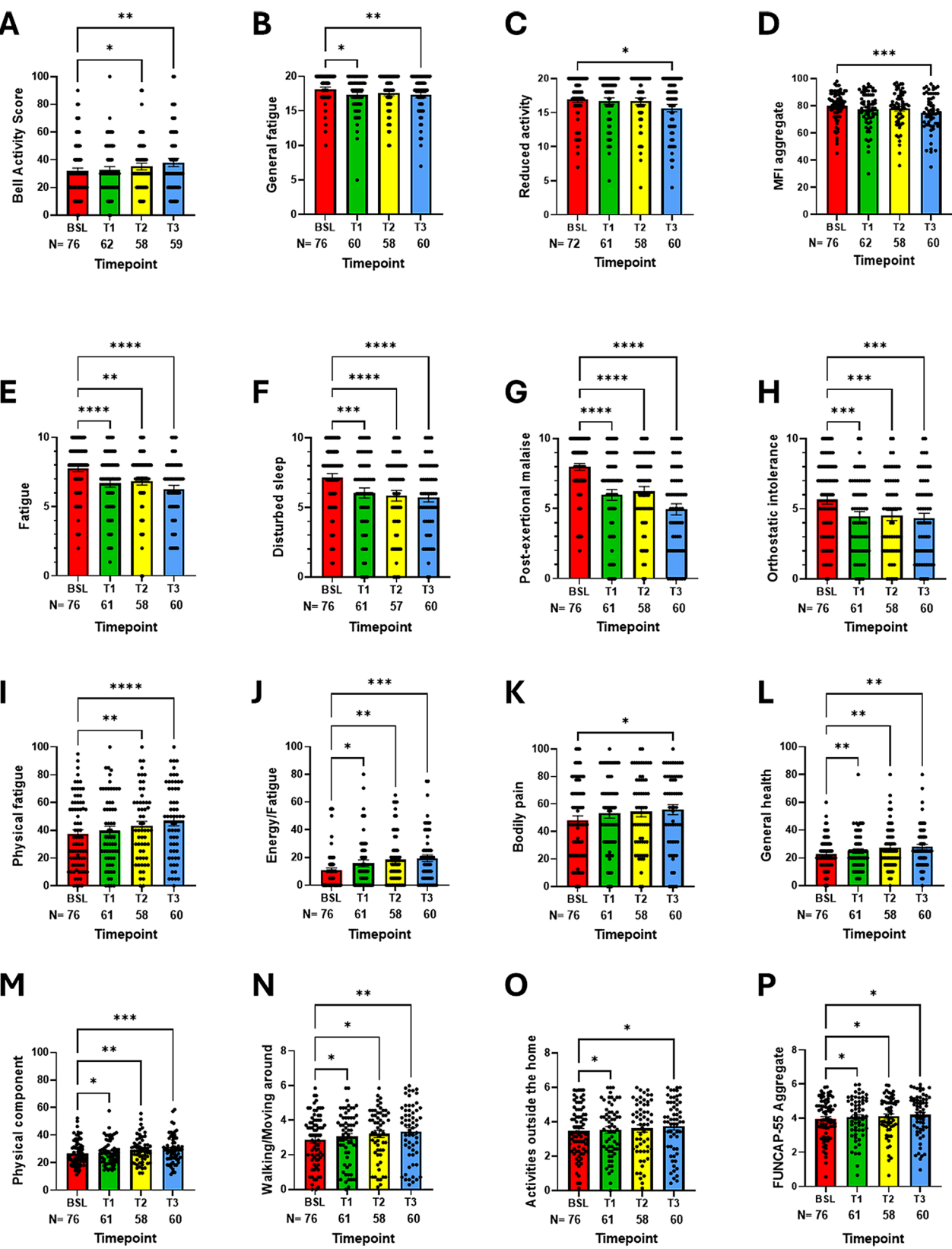


Fig. 1 (See legend on next page.)

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Fig. 1 Compounded version of rapamycin significantly reduces clinical symptoms of general and physical fatigue in ME/CFS Patients. Summary of clinical assessments of physical fatigue scale after 30 and 90 days of rapamycin treatment in N=76 ME/CFS subjects who are considered trial completers and active participants. **(A)** Bell activity scale (BAS) ($^{**}p=0.001$); Multidimensional Fatigue Inventory parameters including **(B)** General fatigue ($^{*}p=0.0021$), **(C)** Reduced Activity ($^{*}p=0.0253$), and **(D)** Multidimensional fatigue inventory (MFI) aggregate ($^{**}p=0.0018$); Specific Symptom Severity (SSS) parameters including **(E)** fatigue ($^{****}p<0.0001$), **(F)** disturbed sleep ($^{****}p<0.0001$), **(G)** post exertional malaise ($^{****}p<0.0001$), **(H)** orthostatic intolerance ($^{****}p<0.0001$); RAND 36-Item Short Form (SF-36). Parameters such as **(I)** physical fatigue ($^{****}p<0.0001$), **(J)** energy/fatigue ($^{****}p<0.0001$), **(K)** bodily pain ($^{*}p=0.0309$), **(L)** general health ($^{***}p=0.0005$), **(M)** physical component score ($^{***}p=0.0002$); and FUNCAP-55 Functional Capacity items such as **(N)** walking/moving around ($^{***}p=0.0003$) **(O)** Activities outside the home ($^{*}p=0.014$), and **(P)** FUNCAP-55 Functional aggregate ($^{*}p=0.0191$) were analyzed and plotted as scatter histogram in ME/CFS subjects at BSL, T1, and T3 stages of trial. Comparisons are from a mixed-effects model. **** : $p < 0.0001$, *** : $p < 0.001$, ** : $p < 0.01$, * : $p < 0.05$

Assessment of general and physical symptoms of fatigue

All analyses were conducted using the combined cohort of trial completers and active participants ($N=76$). Treatment with low-dose rapamycin for 90 days resulted in statistically significant improvements in physical activity as measured by the Bell Activity Scale (BAS) (Fig. 1A). Mixed-effects modeling demonstrated a significant effect of time on BAS scores ($F_{2,36,138} = 6.57$, $^{**}p=0.001$). Post-hoc Dunnett's multiple comparisons testing revealed a significant increase in BAS scores from baseline (32.2) to T3 (38.1) ($^{**}p=0.0055$). Significant improvement was also observed between BSL and T2 ($^{*}p=0.0209$). These findings indicate progressive improvement in physical activity over the 90-day treatment period, with the most robust effects observed at T3.

Rapamycin treatment also significantly improved multiple fatigue-related domains assessed using the Multidimensional Fatigue Inventory (MFI), including general fatigue ($F_{2,63,153} = 5.51$, $^{**}p=0.0021$) (Fig. 1B), reduced activity ($F_{2,84,164} = 3.26$, $^{*}p=0.0253$) (Fig. 1C), and aggregate MFI score ($F_{2,43,144} = 5.90$, $^{**}p=0.0018$) (Fig. 1D). Post-hoc analyses demonstrated significant reductions in general fatigue from BSL to T1 and T3 ($^{*}p=0.0146$, $^{**}p=0.0067$), as well as significant improvement in reduced activity ($^{*}p=0.0389$) and aggregate MFI score ($^{**}p=0.0009$) from BSL to T3.

Physical fatigue showed a strong trend toward improvement but did not reach statistical significance ($F_{2,55,149} = 2.77$, $^{*}p=0.053$; BKY FDR $q=0.0264$) (Supplementary Fig. 6A).

Assessment using the Specific Symptom Severity (SSS) scale demonstrated significant reductions in several hallmark ME/CFS symptoms, including fatigue ($F_{2,73,160} = 15.4$, $^{****}p<0.0001$) (Fig. 1E), disturbed sleep ($F_{2,70,157} = 11.6$, $^{****}p<0.0001$) (Fig. 1F), post-exertional malaise (PEM) ($F_{2,88,169} = 26.0$, $^{****}p<0.0001$) (Fig. 1G), and orthostatic intolerance (OI) ($F_{2,79,163} = 9.49$, $^{****}p<0.0001$)

(Fig. 1H). Post-hoc analyses revealed sustained and statistically significant reductions across all timepoints (T1–T3), indicating durable clinical improvements throughout the treatment period.

Low-dose rapamycin treatment further improved multiple domains of general and physical health assessed by the RAND SF-36 questionnaire. Significant improvements were observed in physical fatigue ($F_{2,43,142} = 12.4$, $^{****}p<0.0001$) (Fig. 1I), energy/fatigue ($F_{2,58,151} = 9.30$, $^{****}p<0.0001$) (Fig. 1J), bodily pain ($F_{2,33,137} = 3.36$, $^{*}p=0.0309$) (Fig. 1K), general health ($F_{2,42,142} = 7.12$, $^{****}p<0.0001$) (Fig. 1L), and aggregate physical component score ($F_{2,42,142} = 8.01$, $^{***}p=0.0002$) (Fig. 1M). Improvements in role limitations due to physical health approached significance but did not survive statistical correction (Supplementary Fig. 6B).

Assessment using the FUNCAP-55 Functional Capacity questionnaire demonstrated significant improvements in walking/mobility ($F_{2,14,125} = 8.46$, $^{***}p=0.0003$) (Fig. 1N), activities outside the home ($F_{2,14,126} = 4.27$, $^{*}p=0.014$) (Fig. 1O), and aggregate FUNCAP-55 score ($F_{0.975,57.2} = 5.88$, $^{*}p=0.0191$) (Fig. 1P). It was also found that there were significant improvements in personal hygiene ($F_{0.435, 25.5} = 3.15$, $p=0.1019$) (BKY FDR $^{*}q=0.0428$) (Supplementary Fig. 6C) from BSL to T3 ($^{*}p=0.044$) and significant improvement in activities outside the home from BSL to T1 and T3 ($^{*}p=0.0246$, $^{*}p=0.0294$). No significant changes were observed in upright time, household activities, or sensory sensitivity to light and sound (Supplementary Fig. 6D–F). Post-hoc analyses demonstrated sustained improvements in mobility and aggregate functional capacity measures across all timepoints.

Assessment of emotional symptoms of fatigue

Mental and emotional components reported in the Multidimensional Fatigue Inventory survey (MFI) showed statistically significant improvements in mental fatigue ($F_{2,91, 170} = 4.82$, $^{**}p=0.0033$) from BSL to T1, T2, and T3 (Fig. 2A). Dunnett's post-hoc test showed significant improvements in mental fatigue T1, T2, and T3 ($^{*}p=0.0234$, $^{*}p=0.0445$, $^{**}p=0.0022$). No statistical significance was observed when evaluating reduced motivation ($F_{2,60, 149} = 2.07$, $p=0.1154$ [ns]) (BKY FDR $^{*}q=0.0441$) (Supplementary Fig. 7A).

Table 4 Adverse events and medication stops timeline

Adverse Event/Reason	RAMP Period	T1 to T2	T2 to T3
GI Complaints/Diarrhea	1	0	0
Muscle Twitching	1	0	0
Tenosynovitis	0	1	0
Erythematous Dermatitis	1	0	0
Total	3	1	0

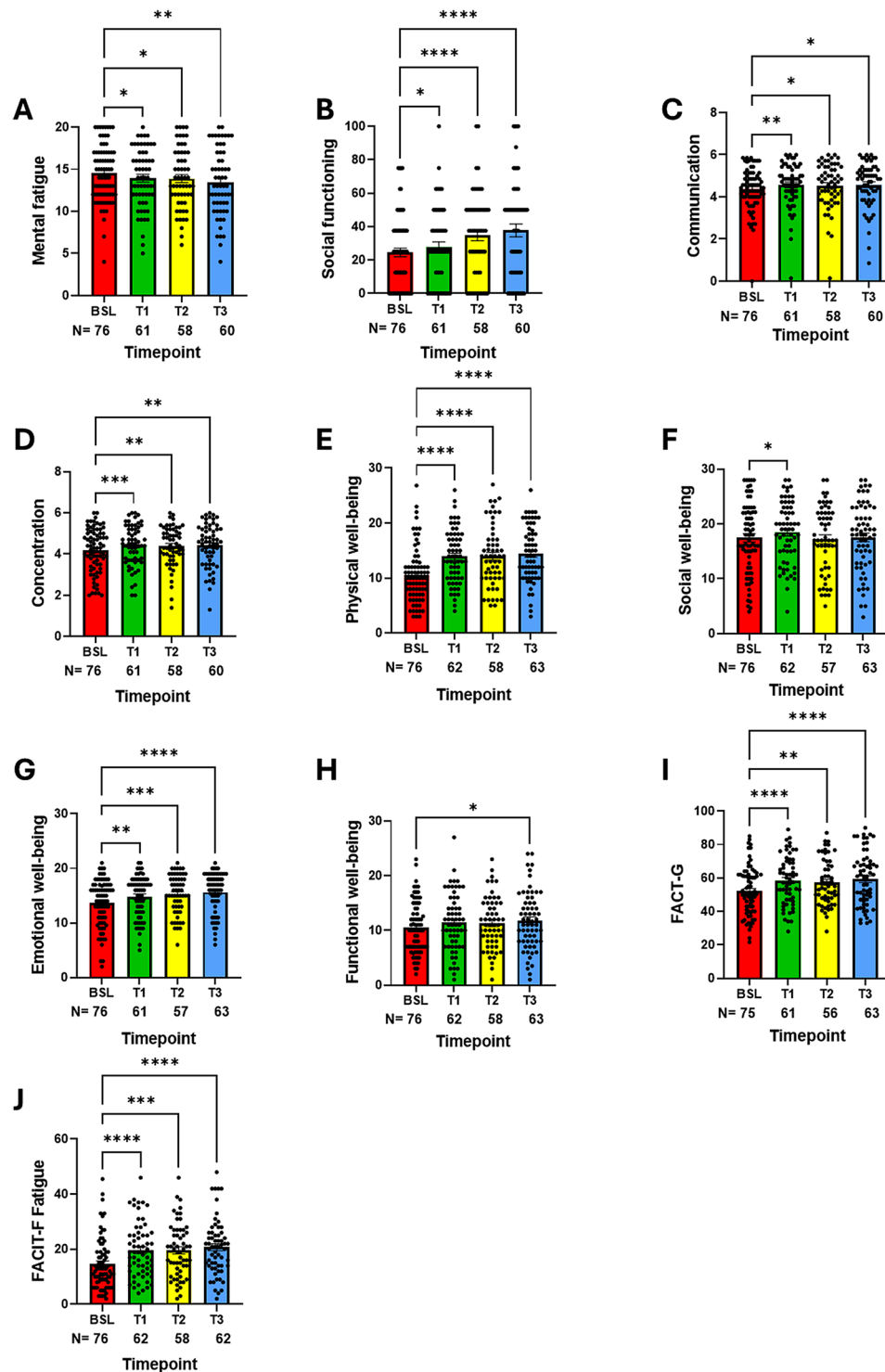


Fig. 2 Compounded version of rapamycin significantly alleviates clinical symptoms of mental or emotional fatigue in ME/CFS Patients. Clinical symptoms for the emotional assessment of fatigue are summarized below (N=76) in ME/CFS trial completers and active participants. MFI parameter (**A**) Mental fatigue (** $p=0.0033$); SF-36 parameter (**B**) social functioning (**** $p<0.0001$); FUNCAP-55 Functional Capacity parameters (**C**) communication (** $p=0.006$) and (**D**) concentration (*** $p=0.0001$); and Functional Assessment of Chronic Illness Treatment (FACIT) parameters (**E**) physical well-being (**** $p<0.0001$), (**F**) social well-being ($p=0.073$ [ns]), (**G**) emotional well-being (**** $p<0.0001$), (**H**) functional well-being ($p=0.111$ [ns]), (**I**) FACT-G (**** $p<0.0001$), and (**J**) FACIT-Fatigue (FACIT-F) (**** $p<0.0001$). Comparisons are from a mixed-effects model. ****: $p < 0.0001$, ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$

Clinical symptoms of social functioning ($F_{2.88, 169} = 12.0$, **** $p < 0.0001$) (Fig. 2B), as measured by SF-36, followed by Dunnett's multiple comparison analysis displayed a significant improvement at T1, T2, and T3 (* $p = 0.0482$, **** $p < 0.0001$, **** $p < 0.0001$). There was no significance observed in role limitations due to emotional problems ($F_{2.40, 141} = 0.511$, $p = 0.6343$ [ns]) (Supplementary Fig. 7B), emotional well-being ($F_{2.07, 122} = 2.47$, $p = 0.0865$) (BKY FDR $q = 0.0376$) (Supplementary Fig. 7C), and the aggregate mental component score ($F_{2.54, 149} = 2.40$, $p = 0.0799$ [ns]) (BKY FDR $q = 0.0360$) (Supplementary Fig. 7D).

PEM-based mental and emotional clinical symptoms assessed by the FUNCAP-55 Functional Capacity questionnaire showed statistically significant improvements specifically in the categories of communication ($F_{1.56, 91.7} = 6.18$, ** $p = 0.006$) (Fig. 2C) and concentration ($F_{2.17, 127} = 9.05$, *** $p = 0.0001$) (Fig. 2D). Post-hoc testing with Dunnett's test showed significant improvements in communication from BSL to T1, T2, and T3 (* $p = 0.001$, * $p = 0.0159$, * $p = 0.0201$). Similar analysis revealed significant improvement in concentration from BSL to T1, T2, and T3 (*** $p = 0.0004$, ** $p = 0.0037$, ** $p = 0.0022$).

Non-canonical functional and quality-of-life symptom measurements

Additional quality-of-life assessments utilizing the Functional Assessment of Chronic Illness Therapy–Fatigue (FACIT-F) and Functional Assessment of Cancer Therapy–General (FACT-G) instruments demonstrated significant improvement across several domains following low-dose rapamycin therapy. Significant improvements were observed in physical well-being ($F_{2.77, 166} = 27.1$, **** $p < 0.0001$) (Fig. 2E), emotional well-being ($F_{2.69, 160} = 12.0$, **** $p < 0.0001$) (Fig. 2G), FACT-G total score ($F_{2.45, 144} = 13.5$, **** $p < 0.0001$) (Fig. 2I), and FACIT-F score ($F_{2.74, 164} = 15.1$, **** $p < 0.0001$) (Fig. 2J). Social well-being (Fig. 2F) and functional well-being (Fig. 2H) demonstrated trends (but not significant) toward improvement that survived BKY false discovery rate correction.

Collectively, these findings indicate that 90-day treatment with low-dose compounded rapamycin significantly improved general, physical, emotional, and cognitive symptoms of ME/CFS while simultaneously enhancing functional capacity and quality-of-life metrics across multiple validated clinical instruments, including BAS, SSS, MFI, SF-36, FUNCAP-55, FACT-G, and FACIT-F.

Assessment of primary endpoints

As mentioned previously, the primary endpoints for this trial included the SF-36, MFI, and BAS. Shown in Table 5 are the quantitative results for these primary endpoints. The SF-36, BAS, and MFI individual and aggregate scores were the primary endpoints. For the SF-36, scores were

normalized to the healthy US adult population, centered at 50, and higher scores indicate better results. For the MFI, lower scores indicate better results. For the BAS, higher scores indicate better results. Significant p-values and q-values (< 0.05) are in bold font.

Assessment of secondary endpoints

The FUNCAP, FACIT, and SSS questionnaires were considered the secondary endpoints for this trial. For the FUNCAP, higher scores indicate lower functional impairment. FACIT-F including FACIT-Fatigue and FACT-G, higher scores indicate less fatigue and a higher quality of life. Finally, for the SSS, lower scores indicate lower symptom severity. Shown in Table 6 are the quantitative results for these secondary endpoints. All significant p and q-values are displayed in bold font.

Rapamycin treatment improves mitochondrial energy metabolism in ME/CFS subjects with PEM

Impaired autophagy resulting from mTOR-mediated ATG13 inactivation [9], together with inflammatory signaling [17], may contribute to persistent muscle fatigue observed in ME/CFS pathogenesis. Our recent clinical study [21] demonstrated that low-dose rapamycin, a selective mTOR inhibitor, alleviated fatigue symptoms in a small patient cohort of ME/CFS subjects. Although restoration of ATG13-dependent autophagy may explain the therapeutic mechanism of rapamycin, the downstream functional consequences on mitochondrial energy metabolism remain incompletely understood. Previous work from our group demonstrated that genetic ablation of ATG13 impairs autophagy and significantly disrupts mitochondrial bioenergetics, drive proinflammatory M1 microglia, augments oxidative stress, and disrupts myelin integrity in muscle serving nerves [17]. Therefore, if mTOR 1 blockade via rapamycin reestablishes autophagy, could these downstream effector pathways also be positively affected via low dose rapamycin?

Given the potential role of rapamycin in restoring autophagic function, we performed Seahorse extracellular flux analyses to evaluate mitochondrial oxidative phosphorylation (OXPHOS) in peripheral blood mononuclear cells (PBMCs) isolated from ME/CFS subjects prior to rapamycin treatment (BSL; $N = 20$), after 30 days of therapy (T1; $N = 10$), and following 90 days of treatment (T3; $N = 8$). To nullify the biasness of the study, we included both rapamycin responders ($N = 9$), partial responders ($N = 13$), and non-responders ($N = 1$) in BSL, T1, and T3 group. There were also two participants with unknown responder status due to dropping out due to an AE ($N = 1$) or an ongoing drug holiday ($N = 1$). To ensure assay consistency, all samples included had approximately 400,000 viable PBMCs per well.

Table 5 Patient questionnaires that served as primary endpoints

Instrument, mean (SEM)	BSL	T1	T2	T3	F	p	q
SF-36							
Physical component	26.6 (1.03)	27.5 (1.15)	29.0 (1.24)	29.8 (1.31)	8.26	<0.001	0.0002
Mental component	38.4 (1.15)	40.2 (1.25)	40.9 (1.25)	41.4 (1.37)	2.35	0.084	0.0360
Physical fatigue	37.6 (2.89)	39.7 (3.31)	42.6 (3.31)	47.1 (3.39)	12.9	<0.001	<0.0001
Role limitations due to:							
Physical problems	4.50 (1.88)	6.15 (2.16)	11.0 (3.38)	11.9 (3.23)	2.64	0.069	0.0290
Emotional problems	46.1 (5.41)	54.1 (5.63)	50.3 (5.78)	52.5 (5.59)	0.44	0.68	0.2220
Energy/fatigue	10.7 (1.64)	15.7 (2.35)	18.1 (2.42)	19.2 (2.42)	9.23	<0.001	0.0002
Emotional well-being	63.0 (2.40)	63.9 (2.62)	66.3 (2.41)	67.4 (2.60)	2.67	0.069	0.0376
Social functioning	24.5 (2.58)	27.9 (2.96)	35.2 (3.29)	37.3 (3.78)	12.1	<0.001	<0.0001
Bodily pain	48.3 (3.16)	53.2 (3.49)	54.9 (3.58)	55.3 (3.58)	3.5	0.025	0.0162
General health	22.8 (1.31)	24.9 (1.70)	27.5 (2.02)	27.7 (1.93)	7.19	0.001	0.0004
MFI							
Aggregate	79.8 (1.26)	77.3 (1.72)	77.8 (1.74)	75.0 (1.77)	5.71	0.002	0.0013
General fatigue	18.2 (0.27)	17.4 (0.36)	17.6 (0.36)	17.3 (0.37)	5.57	0.002	0.0015
Physical fatigue	17.9 (0.30)	17.7 (0.39)	17.5 (0.45)	17.0 (0.46)	2.84	0.049	0.0264
Reduced activity	16.9 (0.38)	16.7 (0.47)	16.7 (0.49)	15.7 (0.53)	3.01	0.034	0.0139
Reduced motivation	12.7 (0.43)	12.2 (0.45)	12.2 (0.50)	11.7 (0.45)	1.78	0.16	0.0441
Mental fatigue	14.6 (0.39)	13.9 (0.45)	13.8 (0.47)	13.4 (0.50)	4.96	0.003	0.0022
BAS	31.8 (2.20)	32.9 (2.33)	35.2 (2.35)	38.1 (2.95)	6.57	0.001	0.0008

Table 6 Patient questionnaires which served as secondary endpoints

Instrument, mean (SEM)	BSL	T1	T2	T3	F	p	q
FUNCAP							
Aggregate	3.97 (0.131)	4.05 (0.153)	4.10 (0.155)	4.19 (0.162)	5.88	0.0191	0.0109
A- Personal Hygiene	4.98 (0.130)	5.02 (0.154)	5.04 (0.149)	5.15 (0.148)	3.15	0.1019	0.0428
B- Walking/Moving Around	2.87 (0.169)	3.05 (0.195)	3.21 (0.196)	3.35 (0.212)	8.46	0.0003	0.0003
C- Being Upright	4.42 (0.145)	4.31 (0.178)	4.48 (0.175)	4.48 (0.180)	0.758	0.4289	0.1544
D- Activities in the Home	3.31 (0.185)	3.37 (0.212)	3.44 (0.213)	3.60 (0.212)	2.19	0.112	0.0441
E- Communication	4.38 (0.113)	4.57 (0.135)	4.54 (0.143)	4.59 (0.140)	6.18	0.006	0.0038
F- Activities Outside the Home	3.46 (0.167)	3.54 (0.192)	3.62 (0.208)	3.75 (0.206)	4.27	0.014	0.0084
G- Reactions to Light and Sound	4.13 (0.142)	4.10 (0.173)	4.13 (0.170)	4.17 (0.174)	1.09	0.3312	0.1227
H- Concentration	4.18 (0.123)	4.43 (0.127)	4.39 (0.128)	4.42 (0.135)	9.05	0.0001	0.0001
FACIT							
FACIT-F	14.7 (1.06)	19.7 (1.25)	19.6 (1.28)	20.7 (1.28)	15.1	<0.0001	<0.0001
FACT-G	52.1 (1.68)	58.5 (1.84)	57.4 (1.86)	59.4 (2.00)	13.5	<0.0001	<0.0001
SSS							
Fatigue	7.76 (0.222)	6.69 (0.290)	6.86 (0.294)	6.25 (0.285)	15.4	<0.0001	<0.0001
Disturbed Sleep	7.16 (0.271)	6.05 (0.367)	5.84 (0.384)	5.73 (0.332)	11.6	<0.0001	<0.0001
Post-Exertional Malaise	7.99 (0.237)	5.98 (0.390)	6.22 (0.358)	4.95 (0.397)	26	<0.0001	<0.0001
Orthostatic Intolerance	5.66 (0.321)	4.44 (0.342)	4.52 (0.370)	4.32 (0.372)	9.49	<0.0001	0.0001

Seahorse OCR analysis (Fig. 3A) demonstrated significant impairment of mitochondrial OXPHOS (Fig. 3B) ($F_{2.667, 236.5}=17.72$; $^{****}p<0.0001$) and reserve respiratory capacity (Fig. 3C) ($F_{2.035, 107.2}=9.312$; $^{**}p<0.0002$) in subjects with ME/CFS compared to healthy controls (HC; $N=10$). Mixed-effects ANOVA further demonstrated that T3 PBMCs exhibited significant restoration of OXPHOS ($^{**}p<0.005$) and reserve capacity ($^{***}p<0.0005$) relative to baseline. Increased OXPHOS reflects enhanced ATP-linked mitochondrial respiration,

whereas increased reserve capacity suggests improved mitochondrial resilience under stress conditions.

To further assess mitochondrial oxidative stress, mitochondrial superoxide production was quantified using the MitoSOX™ assay. Kinetic fluorescence analyses (Fig. 3D), single cell imaging quantification (Fig. 3E), and three-dimensional surface plot analyses demonstrated significantly elevated mitochondrial superoxide levels in PBMCs derived from ME/CFS subjects compared to healthy controls. Collectively, these findings suggest that

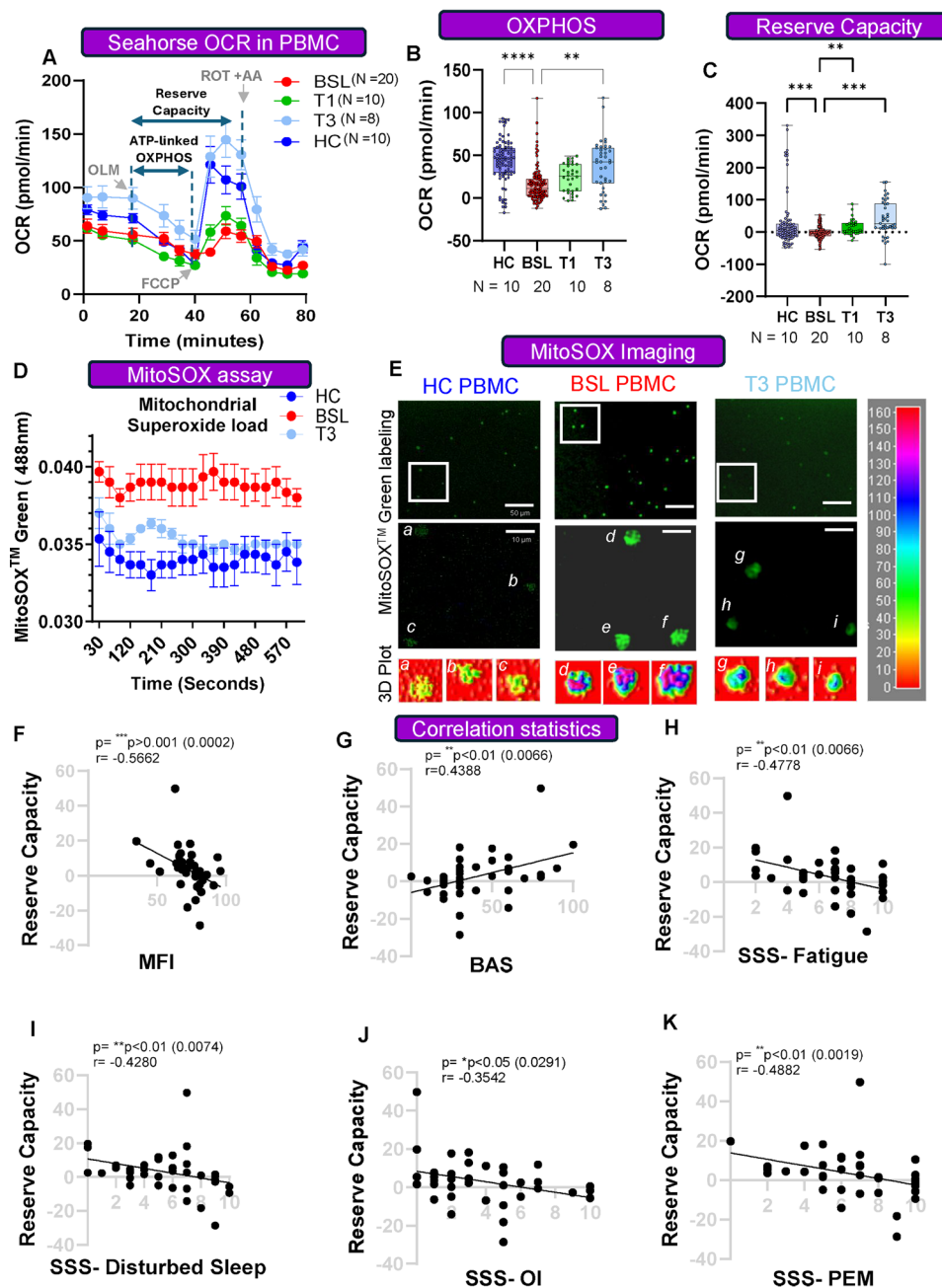


Fig. 3 Mitochondrial respiration analyses in PBMCs of ME/CFS subjects after rapamycin treatment. **(A)** OCR plot measures mitochondrial respiration in PBMCs isolated from HC (deep blue; N=10), and BSL (deep red; N=20), T1 (N=10; green) and T3 (N=8; light blue) ME/CFS subjects. Approximately 400,000 live PBMCs/well were used for the OCR assay. **(B)** Scatter boxplot analyses represented comparisons of **(B)** oxidative phosphorylation (OXPHOS) and **(C)** reserve capacity (RC) among 4 groups. One-way ANOVA combined with Dunnett's multiple comparison analyses were performed to test the significance of mean among groups. Ns = not significant, **p < 0.005, ***p < 0.0005, and ****p < 0.0001. **(D)** PBMCs (~25,000 cells/well; N = 3 HC, BSL, and T3) were assayed for superoxide load by staining with MitoSOXTM green reagent as per manufacturer's instruction followed by a kinetic study at OD@488 nm with 30 sec interval for 10 minutes. Results are mean ± SEM of three different reactions. **(E)** Live cell fluorescence imaging of PBMCs after staining with Mito-SOXTM green. Superoxide-filled mitochondria in each individual cell were further visualized by a 3D surface plot drawn in ImageJ software. Spearman (non-parametric) correlation analyses between mitochondrial reserve capacity and **(F)** MFI or **(G)** BAS or **(H)** fatigue or **(I)** disturbed sleep or **(J)** OI, or **(K)** PEM was conducted in all 38 pairs (20 BSL + 10 T1 + 8 T3) followed by testing significance of correlation by Spearman r and p value displayed on the top of each plot. Results are confirmed after 3 different experiments.

rapamycin therapy may partially restore mitochondrial energy metabolism and reduce oxidative stress in ME/CFS subjects. To determine whether these mitochondrial improvements correlated with clinical recovery, correlation analyses between reserve capacity and multiple fatigue-related scoring method such as MFI (Fig. 3F), BAS (Fig. 3G), Fatigue (Fig. 3H), Disturbed Sleep (Fig. 3I), OI (Fig. 3J), and PEM (Fig. 3K). Significant correlations were observed across all analyses ($\alpha=0.05$), suggesting that improved mitochondrial reserve capacity is strongly associated with amelioration of fatigue-related symptoms following 90 days of rapamycin treatment. Similar correlation analyses involving OXPHOS did not yield statistically significant results (Supplementary Fig. 8). Overall, these correlations indicate that rapamycin-associated restoration of mitochondrial stress resilience may contribute to clinical improvement in ME/CFS subjects who report a positive response to rapamycin therapy.

Rapamycin modulates purine biosynthesis through regulation of IMPDH2 activity

Our next objective was to investigate the molecular mechanisms through which rapamycin restores mitochondrial energy metabolism in ME/CFS subjects. Recent studies [35, 38] suggest that disturbances in purine metabolism may contribute to ME/CFS pathogenesis by altering circulating purine bioavailability. We previously demonstrated [35, 45] that impaired glycolytic and mitochondrial energy metabolism may drive compensatory activation of the pentose phosphate pathway, thereby enhancing purine biosynthesis.

Measurements of plasma purines

To examine the relationship between purine metabolism and rapamycin-mediated metabolic recovery, we performed targeted LC-MS metabolomic analyses of plasma purines in ME/CFS subjects before and after rapamycin treatment. Method optimization, chromatographic calibration, standard curve generation, and metabolite characterization were performed using our in-house metabolomics platform as described comprehensively in the Methods section (Supplementary Fig. 9).

Specifically, plasma levels of inosine monophosphate (IMP), xanthosine monophosphate (XMP), and hypoxanthine (HPX) in ME/CFS subjects ($N=9$) before treatment (BSL) and after 90 days of rapamycin therapy (T3) (Fig. 4A) were analyzed. These individuals demonstrated significant clinical improvement based on BAS (Fig. 4B) and MFI (Fig. 4C) scores. LC-MS analyses revealed significant upregulation of IMP following rapamycin treatment, whereas XMP level was markedly reduced. These changes were confirmed by both pairwise (Fig. 4D–E) and cumulative statistical analyses (Fig. 4F–G).

Are these detected levels of IMP and XMP physiologically relevant? Using PPB to μM conversion formula ($\mu\text{M} = \text{PPB} / \text{molar mass}$), average levels ($\pm$ SD) of IMP (molar mass = 348.21 $\mu\text{g}/\text{mL}$) is 0.956 ± 0.704 in BSL and 1.83 ± 0.442 μM (Fig. 4H). Even though this is a very low-abundant purine metabolite, this range matches with previously reported data by other groups [47, 48]. Using similar conversion formula, XMP levels (molar mass = 364.21 g/mol) were found 0.134 ± 0.086 in BSL and 0.0912 ± 0.070 in T3 plasma. Since that level of XMP is very close to not-detected level the quantitative estimation of XMP in plasma may not be reliable (Fig. 4H). However, circulating levels of XMP have also been reported by other groups [49, 50], and their findings fall within a similar concentration range. Furthermore, to better understand the metabolic regulation of IMP to XMP conversion, PBMC-based cellular assay such as IMPDH kinase activity and FACS analysis needs to be done.

HPX, a downstream purine intermediate, was also found to be significantly decreased after rapamycin treatment (Fig. 4I). Reduced IMP levels at baseline may promote increased conversion toward HPX [51], which is known to contribute to peroxide radical generation and oxidative stress through xanthine oxidase-mediated metabolism [39, 52] (Fig. 4J). Together, these findings suggest that rapamycin treatment may alter purine flux toward a less oxidative and less inflammatory metabolic profile.

IMPDH enzymic activity and IMPDH2-Active PBMC population analyses by FACS

Both IMP and XMP function as substrate and product, respectively, in a purine biosynthetic pathway catalyzed by inosine monophosphate dehydrogenase (IMPDH) (Fig. 4J). Although PBMCs express both IMPDH1 and IMPDH2 isoforms [53], IMPDH2 is preferentially regulated during inflammatory activation [54]. The reciprocal regulation between IMP and XMP following rapamycin treatment suggests that IMPDH2 activity may be altered in treated subjects who reported a positive response to rapamycin therapy.

To evaluate IMPDH2 activity, flow cytometric analyses were performed in PBMCs isolated from six ME/CFS subjects at baseline and T3. Pseudo color density plot (Fig. 4K–L) and histogram analyses (Fig. 4M) demonstrated a distinct population of IMPDH2-immunoreactive PBMCs in baseline ME/CFS samples, which was markedly attenuated following rapamycin treatment at T3. These findings suggest that rapamycin suppresses IMPDH2 activity in circulating immune cells. In order to apply more direct approach monitoring IMPDH activity (Fig. 4N), we performed a kinetic-based enzyme activity assay in PBMCs of ME/CFS subjects ($N=6$) before

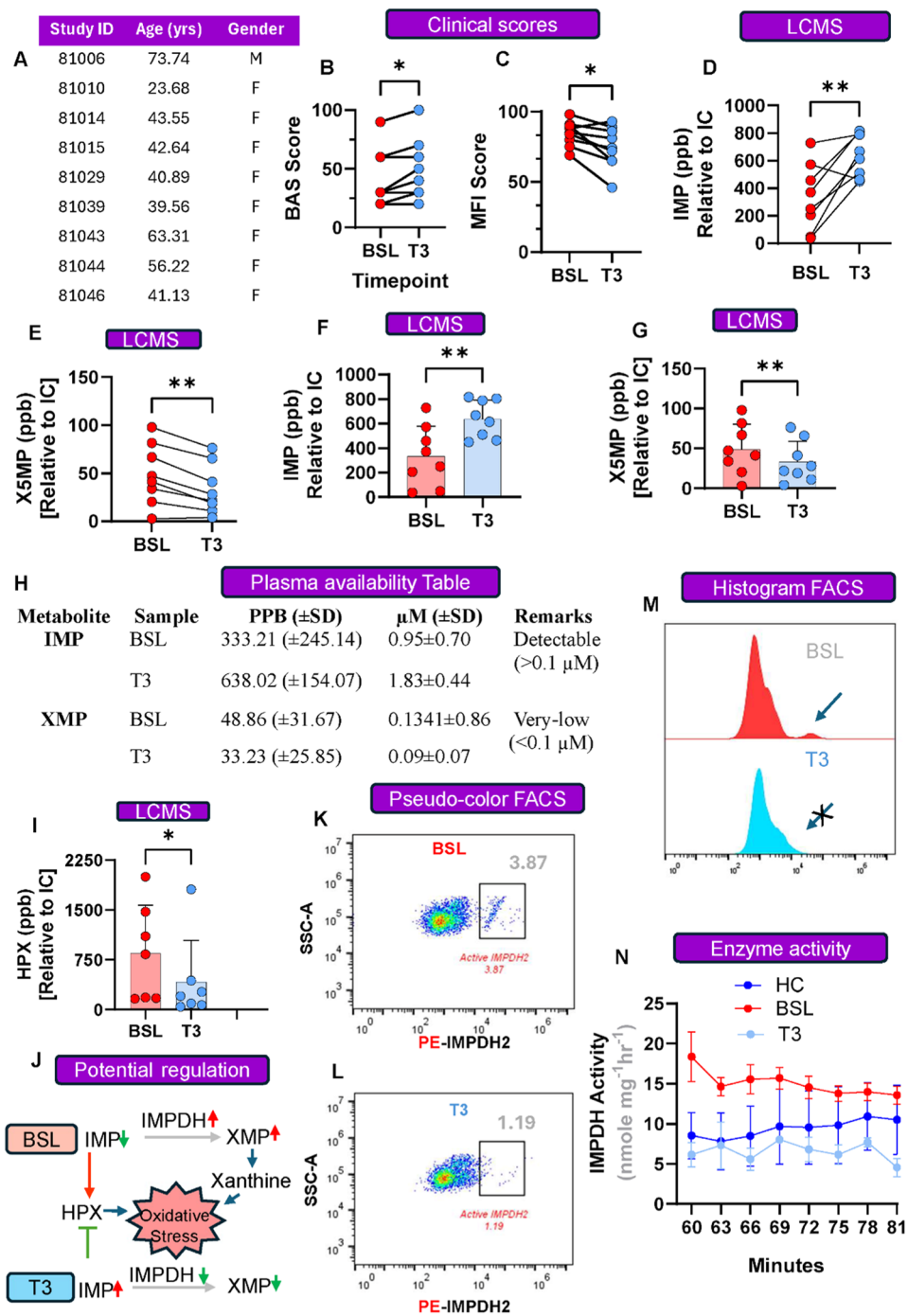


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Fig. 4 Rapamycin treatment regulates the bioavailability of purine metabolites in the plasma samples of ME/CFS subjects. **(A)** The table represents a responder cohort (N=9/gr), including their ages and genders, for a pair-wise analysis of purine metabolites via LCMS before (BSL) and 90 days after (T3) rapamycin treatment. **(B)** Pair-wise comparison of BAS score between BSL and T3 subjects was assessed by a scatter plot analysis followed by a paired t-test measuring the significance of mean, which results $^*p<0.05$ ($=0.0222$); $t=2.827$, $df=8$. **(C)** A scatter plot represents a pair-wise comparison of MFI score followed by a paired t-test to measure the significance. $^*p<0.05$ ($=0.0132$); $t=3.169$, $df=8$. LCMS-derived quantification study was summarized in the pair-wise scatter plots for the relative levels of **(D)** IMP (paired t-test; $^*p<0.05=0.0174$; $t=2.989$; $df=8$; $N=9$) and **(E)** XMP (paired t-test; $^{**}p<0.005=0.0016$; $t=4.671$, $df=8$; $N=9$). Cumulative scatter bar plots summarized the overall LCMS results of **(F)** IMP (unpaired t-test; $^*p<0.05=0.0324$; $t=2.343$, $df=16$; $N=9$), **(G)** XMP (unpaired t-test; $^*p<0.05=0.0498$; $t=2.122$; $df=16$; $N=9$). **(H)** The table demonstrated the conversion from ppb to μM units for both IMP and XMP. The limitation of LCMS-based assay to detect XMP level was identified. **(I)** HPX (t-test; $^*p<0.05=0.0470$). **(J)** A schema to summarize metabolic events that may be responsible for the relative bioavailability of IMP, XMP, and HPX. Flow-cytometry analysis followed by a pseudo-color plot to summarize the active population of IMPDH2-ir cells in PBMCs of **(K)** BSL and **(L)** T3 ME/CFS subjects. **(M)** flowcytometry followed by a histogram representation of IMPDH2 populations in PBMCs. Arrowhead indicates the active IMPDH2 population. **(N)** IMPDH enzyme activity was assessed in HC, BSL, and T3 PBMCs. A 2-way-ANOVA followed by Tukey's multiple comparison analysis revealed that there was a significant upregulation of IMPDH activity in BSL PBMCs compared to T3 [$^{****}p<0.0001$] and HC [$^{**}p<0.002$]. Results are mean $\pm$ SEM of three different experiments

(BSL) and after (T3) rapamycin treatment. The IMPDH activity assay revealed that BSL PBMCs are associated with elevated enzyme activity compared to PBMCs of healthy controls (HCs), whereas that IMPDH activity was found to be normalized in T3 PBMCs corroborating that rapamycin treatment may be associated with the inhibition of upscaled IMPDH activity in ME/CFS subjects.

Recent studies [55, 56] demonstrate that serine phosphorylation regulates both IMPDH1 (Ser160) and IMPDH2 (Ser122) activity. Although these phospho-specific IMPDH antibodies are unavailable, dual flow cytometric (FACS) analyses using anti-IMPDH2 and pan-phosphoserine antibodies (Fig. 5A–B) demonstrated elevated serine-phosphorylated IMPDH2 levels in baseline ME/CFS PBMCs relative to healthy controls (Fig. 5Ci). In contrast, T3 PBMCs exhibited significantly reduced phosphoserine-IMPDH2 staining, suggesting that rapamycin suppresses IMPDH2 activity by reducing its serine phosphorylation state.

Because mTOR functions as a serine kinase and is directly inhibited by rapamycin, we next evaluated mTOR activation status. Flow cytometric analyses demonstrated significantly reduced phospho-mTOR (Ser2448) levels in T3 PBMCs compared to baseline samples (Fig. 5Cii–iii), suggesting that rapamycin-mediated inhibition of mTOR signaling may reduce IMPDH2 phosphorylation and activity.

Purine supplementations regulated ROS production in PBMCs

To further investigate the relationship between purine metabolites and oxidative stress, mitochondrial superoxide levels were measured using MitoSOX-based flow cytometry. Baseline plasma strongly induced mitochondrial superoxide accumulation in PBMCs (Fig. 5D). Supplementation of baseline plasma with XMP sustained mitochondrial ROS production. However, addition of XMP to T3 plasma markedly elevated mitochondrial superoxide generation, whereas IMP supplementation had minimal effect (Fig. 5E). HPX supplementation strongly induced mitochondrial superoxide

in both baseline and T3 PBMCs. These findings suggest that XMP and HPX, downstream products of IMPDH2-mediated purine metabolism, may directly contribute to oxidative stress in ME/CFS PBMCs.

Collectively, these analyses demonstrate that rapamycin suppresses mTOR signaling, decreases IMPDH2 phosphorylation and activity, reduces conversion of IMP to XMP, and lowers the bioavailability of oxidative purine metabolites such as XMP and HPX. These metabolic alterations may contribute to reduced oxidative stress and improved mitochondrial function in ME/CFS subjects.

Rapamycin enhances glycolytic and mitochondrial energy metabolism in myeloid cells

Deficits in glycolysis and mitochondrial oxidative phosphorylation are closely associated with redox imbalance [57, 58], which in turn may promote inflammatory activation of myeloid cells [59]. Because rapamycin treatment reduced mitochondrial superoxide loads accumulation while modulating IMPDH2-dependent purine metabolism synthesis, we next investigated whether altered purine metabolism could influence the energy metabolism in myeloid cells.

Using our previously established hypoxia-induced microglial cell culture model [35], which induces IMPDH2 activity and purine biosynthesis following stimulation with ribose-5-phosphate (R5P), we evaluated the effects of ME/CFS plasma on cellular bioenergetics. In the current study, 10% plasma from ME/CFS subjects was substituted for R5P. Exposure to baseline plasma significantly impaired mitochondrial respiration in C20 microglial cells compared to healthy control plasma, reducing both maximal respiration and reserve respiratory capacity (Supplementary Fig. 10A–C). In contrast, treatment with T3 plasma partially restored reserve respiratory capacity ($^*p<0.05$), although maximal respiration ($^{ns}p>0.05$) remained impaired (Supplementary Fig. 10D–F).

Evaluation of glycolytic metabolism by Seahorse ECAR demonstrated that baseline glycolysis was not

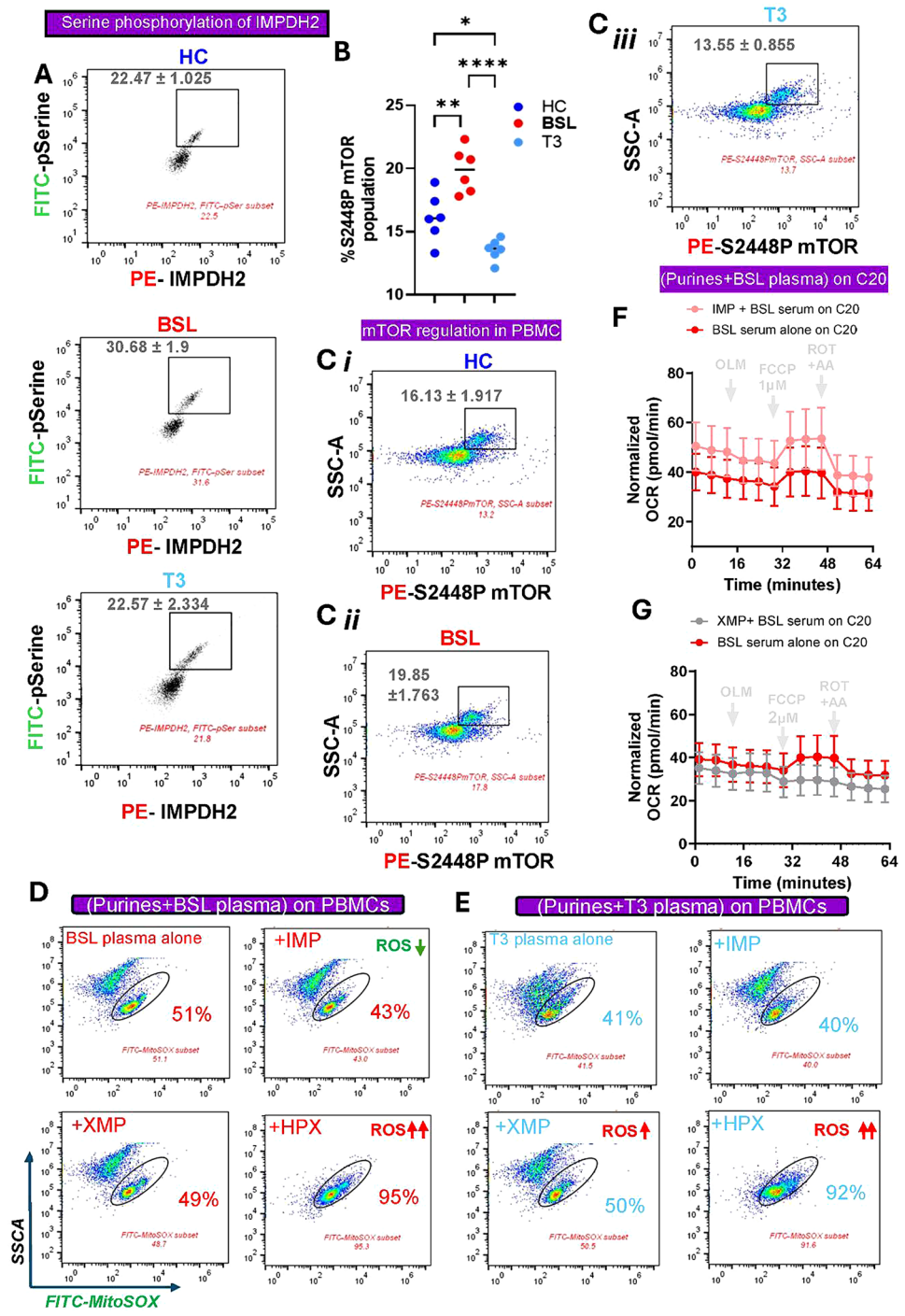


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Fig. 5 Rapamycin mitigates the mitochondrial superoxide level and may improve mitochondrial oxidative metabolism in PBMCs by attenuating IMPDH2 activity via mTOR inhibition. **(A)** Peripheral blood mononuclear cells (PBMCs; 50,000 per sample, N=6/group) from healthy controls (HC), baseline (BSL), and T3-treated groups were assessed for viability, stained with FITC-conjugated pSeriine (ThermoFisher; 1:100 dilution) and PE-conjugated IMPDH2 (ThermoFisher; 1:100 dilution) at room temperature for 1 hour in FACS buffer, washed three times, and analyzed using a BD flow cytometer. Data analysis was performed using FlowJo 10 software, and results are presented as dotted scatterplots (gated cells: 7,000). **(B)** The population enclosed within the defined gate was quantified and displayed as a scatter histogram. Statistical significance of mean differences was determined by one-way ANOVA followed by multiple comparison tests ($p < 0.001$, $nsp > 0.05$). **(C)** Single-color flow cytometry analysis of PBMCs using PE-tagged S2448P mTOR was conducted for (i) HC, (ii) BSL, and (iii) T3-treated groups, with mean $\pm$ SD data provided in the inset. **(D)** BSL PBMCs (N=3/group) were incubated with 10% BSL plasma (v/v), either alone or supplemented with 20 μ M IMP, XMP, or HPX for 2 hours at 37°C in a 5% CO₂ incubator, followed by MitoSOX™ Green staining (ThermoFisher) for 30 minutes prior to flow cytometry imaging (gate: 10,000). **(E)** T3 PBMCs (N=3/group) were treated similarly with T3 plasma or supplemented with different purines and then subjected to the MitoSOX™ Green assay. Seahorse OCR analysis was performed on C20 microglia treated with 10% BSL plasma alone (N=9/gr) or in combination with 20 μ M **(F)** IMP or **(G)** XMP, after incubation under hypoxic conditions, to assess mitochondrial respiration using the Seahorse XF96 analyzer

significantly altered by BSL treatment. However, BSL plasma significantly reduced both glycolytic capacity and glycolytic reserve in microglial cells relative to healthy control plasma (Supplementary Fig. 10G–I). Conversely, treatment with T3 plasma significantly restored glycolytic capacity (Supplementary Fig. 10J–K), suggesting partial recovery of stress-induced glycolytic adaptation following rapamycin therapy. To directly assess the role of purines, C20 cells were treated with BSL plasma supplemented with either IMP or XMP. IMP supplementation significantly (N=9/group) improved the maximal respiration (Fig. 5F) ($^{****}p < 0.0001$), whereas XMP supplementation failed to restore the mitochondrial respiration under either basal or stress-induced conditions (Fig. 5G). These findings support the concept that dysregulated IMPDH2 activity and downstream purine metabolism may contribute to impaired mitochondrial respiration in ME/CFS.

Collectively, these results indicate that baseline ME/CFS plasma primarily impairs stress-adaptive glycolytic and mitochondrial responders rather than basal energy metabolism, whereas plasma obtained after rapamycin treatment partially restores these adaptive metabolic responses in myeloid cells.

Rapamycin reduces pro-inflammatory myeloid polarization in ME/CFS

Impaired oxidative phosphorylation and mitochondrial dysfunction are known to promote the polarization of myeloid cells toward a pro-inflammatory M1 phenotype [17, 60, 61]. To determine whether rapamycin-associated metabolic changes influence inflammatory activation, we utilized our previously established hypoxia-induced microglial polarization model (Fig. 6A).

Flow cytometric analyses demonstrated that baseline ME/CFS plasma significantly increased the CD11b⁺CD40⁺ M1 microglial population relative to healthy control plasma (Fig. 6Bi–ii). In contrast, T3 plasma markedly reduced M1 polarization (Fig. 6Biii), suggesting that rapamycin treatment attenuates inflammatory activation in myeloid cells (Fig. 6C). Similar

findings were observed for the additional M1 marker CD86 (Fig. 6D–E).

Immunofluorescence analyses further confirmed increased CD40 and CD86 expression following exposure to baseline plasma, whereas T3 plasma significantly reduced expression of both markers (Fig. 6F–G). Expression of inducible nitric oxide synthase (iNOS), a functional marker of inflammatory M1 myeloid cells [62], was also strongly induced by baseline plasma and attenuated following treatment with T3 plasma (Supplementary Fig. 11B). No significant changes were observed in arginase expression (Supplementary Fig. 11A).

We next evaluated anti-inflammatory M2 polarization by assessing CD206 expression. Flow cytometric analyses demonstrated that T3 plasma induced a distinct CD11b⁺CD206⁺ population (Fig. 6H–I), suggesting that rapamycin-associated plasma factors may promote anti-inflammatory myeloid differentiation.

Overall, these findings indicate that rapamycin treatment may reduce inflammatory activation by suppressing M1 polarization while promoting M2-associated phenotypes in myeloid cells.

Purinergic receptor signaling differentially regulates myeloid bioenergetics and inflammation in ME/CFS

Because rapamycin substantially altered purine metabolism through regulation of IMPDH2 activity, we next investigated whether circulating purines contribute to inflammatory activation through purinergic receptor signaling pathways. Myeloid cells are highly responsive to extracellular purines due to their abundant expression of purinergic receptors [63–65].

We focused specifically on the P₂X₄ and P₂X₇ receptors, both previously implicated in inflammatory microglial activation. Flow cytometric analyses confirmed robust expression of both receptors in C20 microglial cells under hypoxic conditions (Supplementary Fig. 12A–B). Treatment with 5 pmoles of receptor-specific siRNA significantly reduced receptor expression without affecting cell viability (Supplementary Fig. 12C).

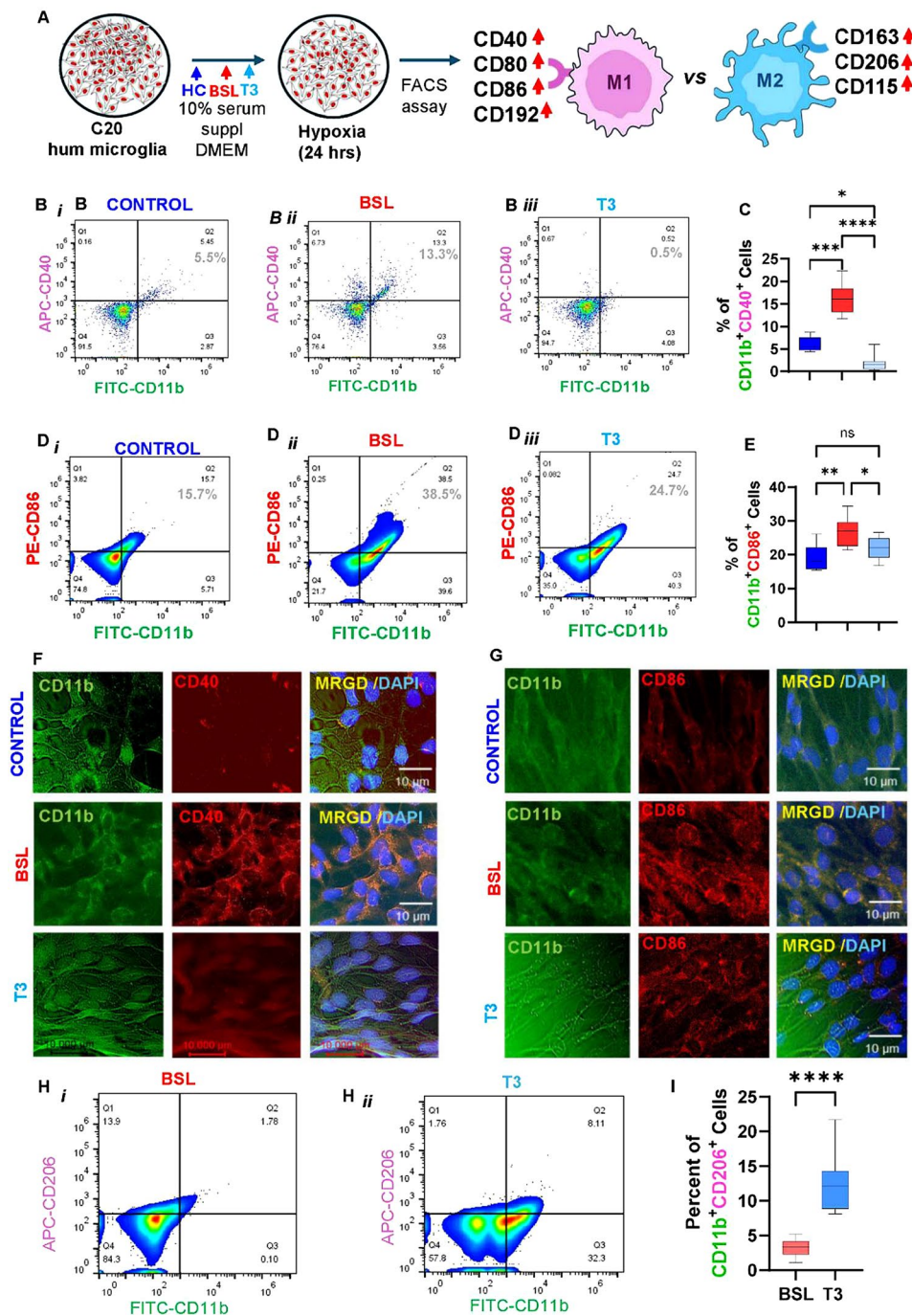


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Fig. 6 Plasma samples of rapamycin-treated ME/CFS subjects suppress the M1 inflammatory phenotype of microglia cells. **(A)** A schema demonstrates an in vitro cell culture strategy to study the effect of ME/CFS plasma samples on the inflammation of microglial cells. C20 microglial cells were treated with N=6 plasma samples (10% v/v in DMEM serum-free media) of HC, BSL, and T3 (responders) ME/CFS subjects, followed by incubation in a hypoxia chamber for 24 hrs. After that, flow-cytometry assays were performed to evaluate surface antigens for M1 versus M2 myeloid cells. **(B)** Dual flow-cytometry analyses in (i) HC, (ii) BSL, and (iii) T3 plasma-treated C20 microglia with pan myeloid marker CD11b (FITC-tagged; 488 nm filter) and M1-specific marker CD40 (APC-tagged; 640 nm filter). Representative images are 4-quadrant pseudo-color plot derived after analyzing 10,000 events at FlowJo V10 software. **(C)** Box plot represents the quantitative estimation of CD11b+CD40+ microglial cells after plasma treatment (N=6 /group). Significance of mean was tested by multiple comparisons following one-way ANOVA. * $p < 0.05$; *** $p < 0.001$; and **** $p < 0.0001$. **(D)** Flow cytometry analyses of pan-myeloid marker CD11b and M1 marker CD86 (PE-tagged; 535 nm filter) in C20 myeloid cells treated with (i) HC, (ii) BSL, and (iii) T3 plasma and then incubated in a hypoxia chamber for 24 hrs. Representative images are smooth pseud-color representations derived from FlowJo V10 software with 10,000 events/group. **(E)** Box plot quantification analyses were conducted to quantify the percentage of CD11b+CD86+ microglial cells. One-way ANOVA followed by multiple comparisons was performed to monitor the significance of the mean between groups. Ns = no significance; * $p < 0.05$, ** $p < 0.01$. Dual IF analyses of **(F)** CD11b /CD40 and **(G)** CD11b/CD86 in HC, BSL, and T3 plasma-treated C20 microglial cells. **(H)** Dual flow-cytometry analyses of CD11b and M2-specific marker CD206 in (i) BSL and (ii) T3 plasma-treated microglial cells. **(I)** Box plot analyses to quantify the percentage of CD11b+CD206+ microglial cells from N=6 different flow cytometry analyses. **** $p < 0.0001$ as confirmed by Dunnett's multiple comparison after one-way ANOVA. Results are mean $\pm$ SD of three different experiments

OCR analyses demonstrated that silencing $P_2 X_4$, but not $P_2 X_7$, significantly restored reserve respiratory capacity in microglial cells exposed to baseline ME/CFS plasma (Fig. 7A–B). In contrast, maximal respiration was only modestly affected (Fig. 7C). These findings suggest that $P_2 X_4$ signaling specifically contributes to stress-induced suppression of mitochondrial energy metabolism. A recent study showed that ME/CFS serum samples did not affect mitochondrial oxidative metabolism in myoblasts [66]. However, myeloid cells express a wider range of inflammatory and purinergic receptors, respond vigorously to hypoxia and stress, and may behave differently.

ECAR analyses further demonstrated that silencing $P_2 X_7$ moderately enhanced glycolytic capacity under stress conditions without affecting basal glycolysis or glycolytic reserve (Fig. 7D–G). Conversely, $P_2 X_4$ silencing significantly reduced basal glycolysis but did not alter glycolytic reserve or maximal glycolytic capacity (Fig. 7H–K). These findings suggest that $P_2 X_4$ and $P_2 X_7$ regulate distinct aspects of myeloid bioenergetics in response to ME/CFS plasma.

Finally, dual flow cytometric analyses demonstrated that silencing either $P_2 X_4$ or $P_2 X_7$ suppressed the induction of the M1 marker CD40 in microglial cells exposed to baseline ME/CFS plasma (Fig. 7L). These findings suggest that purinergic receptor-mediated disruption of glycolytic and mitochondrial metabolism may contribute directly to inflammatory polarization of myeloid cells. Overall, our phase II, non-placebo-controlled, observational study provides evidence that 90 days of low-dose compounded rapamycin treatment alleviates self-reported symptoms reduced fatigue in ME/CFS subjects, including PEM, while simultaneously improving mitochondrial stress resilience, modulating purine metabolism through IMPDH2 regulation, reducing oxidative stress, and attenuating inflammatory activation of myeloid cells. These findings identify purinergic signaling and metabolic dysfunction as potential mechanistic

drivers of ME/CFS pathogenesis and suggest that mTOR inhibition may represent a viable therapeutic strategy to restore immune-metabolic homeostasis in subsets of ME/CFS patients.

Discussion

While rapamycin is by no means a novel therapeutic agent, the use of low-dose regimens, such as those utilized in the present study, differs substantially from the higher doses historically employed in transplantation medicine for immunosuppression. Interest in low-dose rapamycin outside of the clinical context of transplant emerged largely from in vivo studies demonstrating lifespan extension and improved health span in mammalian aging models and now in human studies [67, 68]. Chronic activation of mTOR not only inhibits autophagy but also promotes geroconversion, a transition from transient cell-cycle arrest to a senescent state characterized by cellular hypertrophy, altered metabolic programming, mitochondrial dysfunction, and the development of the senescence-associated secretory phenotype (SASP) [69]. Notably, many of these same biological pathways, including oxidative stress, mitochondrial dysfunction, impaired energy metabolism, and immune dysregulation, are known to be dysfunctional in ME/CFS and dependent on chronic mTOR activation which has been described in a subset of ME/CFS patients. Therefore, clinical trials aimed to negatively modulate mTOR are highly relevant to ME/CFS. This phase II observational trial of rapamycin provides additional support for the clinical utility of rapamycin in the management of key disease-defining symptoms of ME/CFS, particularly post-exertional malaise (PEM), fatigue, and cognitive dysfunction.

Dosing strategy, clinical outcomes, and safety profile in the Phase II trial

Two major differences distinguished the Phase II study from our earlier Phase I trial and may account for the improved retention and reduced frequency of adverse

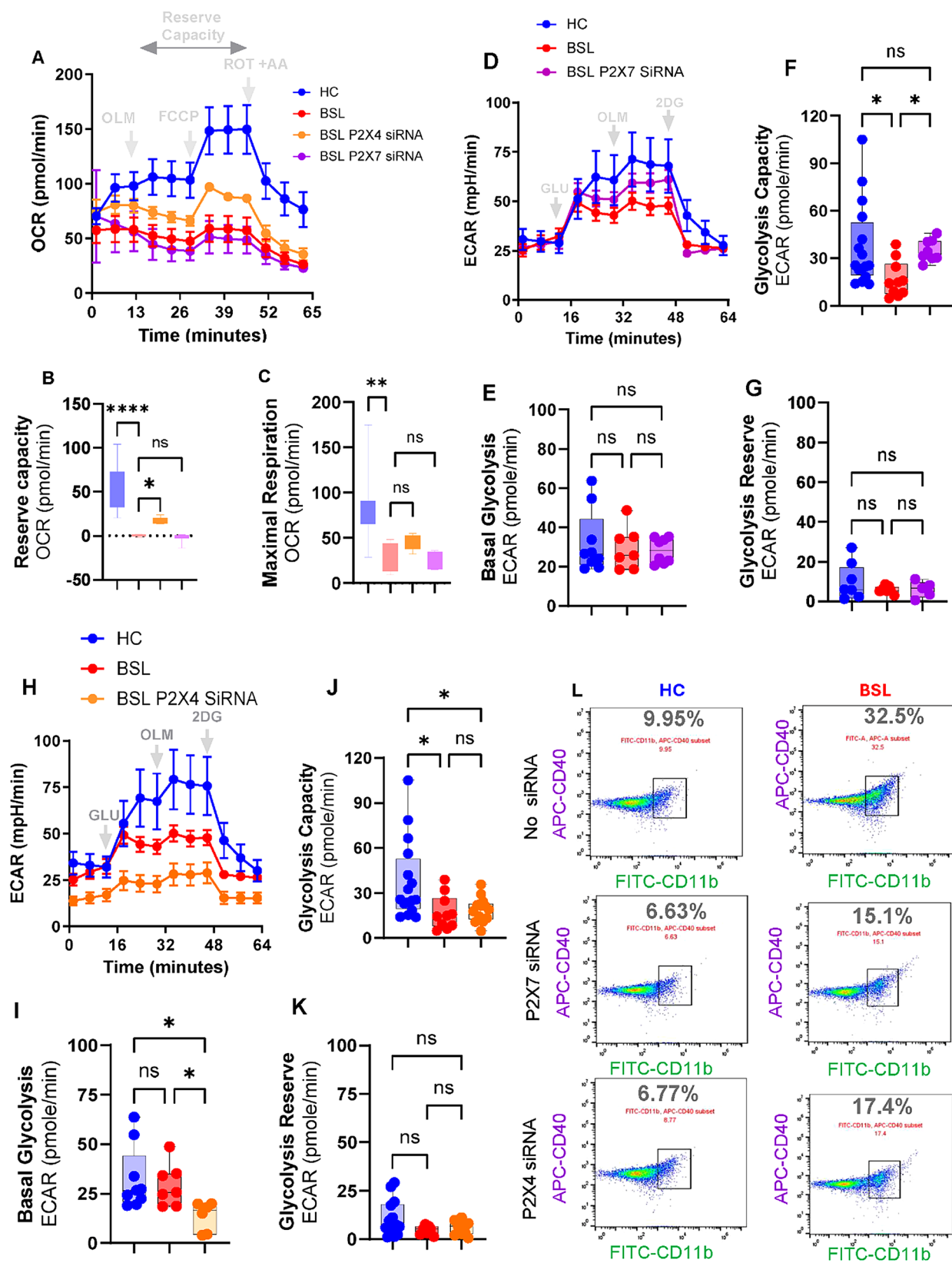


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Fig. 7 Inflammatory M1 phenotype in ME/CFS plasma-treated C20 myeloid cells is regulated by differential purinergic signaling through P_2X_4 and P_2X_7 receptors. **(A)** OCR analysis in C20 microglia silenced with P_2X_4 (orange) or P_2X_7 (purple) for 24 hrs, then treated with BSL (red) (N=8) or HC plasma (blue) for another 24 hrs in a hypoxia chamber and then analyzed for mitochondrial respiration in Agilent XF96 analyzer. Measuring **(B)** reserve capacity and **(C)** maximal respiration were analyzed and compared by Kruskal-Wallis test (**** $p<0.001$, ** $p<0.01$, * $p<0.05$). **(D)** ECAR plot measures glycolysis in C20 microglia treated with HC (deep blue; N=8), and BSL (deep red; N=8) plasma samples and then compared with BSL-treated C20 microglia transfected with P_2X_7 siRNA (purple; N=8). Plasma-treated C20 microglial cells were maintained under hypoxia for 24 hrs before analyzing at Seahorse analyzer for glycolysis. Satter boxplot analyses to compare **(E)** basal glycolysis, **(F)** glycolytic capacity, and **(G)** glycolytic reserve in HC (blue box), BSL (red box), and BSL+ P_2X_7 siRNA (purple box) groups. Since the data distribution in HC did not pass normality test, the Kruskal-Wallis test combined with Dunn's multiple comparison analyses were performed in measuring the significance of mean between groups. Ns = not significant, * $p<0.05$. **(H)** ECAR analyses in C20 microglia transfected with P_2X_4 siRNA followed by the treatment with BSL plasma (orange; N=8 and then compared with the results from C20 microglia treated with either HC (N=8; blue) or BSL plasma alone (red; N=8). After plasma treatment C20 microglia was kept in a hypoxia chamber for 24 hrs before subject to ECAR study. Satter boxplot analyses to compare **(I)** basal glycolysis, **(J)** glycolytic capacity, and **(K)** glycolytic reserve in HC (blue box), BSL (red box), and BSL+ P_2X_4 siRNA (orange box) groups. Ns = not significant, * $p<0.05$ and ** $p<0.01$ as derived by Dunn's multiple comparison after non-parametric Kruskal-Wallis test. Results are mean $\pm$ SEM of three independent experiments. **(L)** Pseudo-color plots represent dual flow cytometry analyses of CD11b and CD40 in C20 microglia cells transfected with no siRNA, P_2X_7 siRNA, and P_2X_4 siRNA followed by incubation in a hypoxia chamber with HC, BSL, and T3 plasma samples. Results are confirmed after 3 different experiments

events. First, phase II utilized a single compounded low-dose rapamycin formulation rather than multiple generic formulations. Although the relative contribution of these factors cannot be independently determined, the marked reduction in adverse events, and the ability to shorten the dose-escalation period from 6 weeks to 4 weeks, suggest that the compounded formulation may be well suited for this patient population, which is characterized by high rates of chemical sensitivity and medication intolerance. Second, several strategies were implemented to improve decentralized trial coordination, including enhanced patient follow-up and increased budgetary resources, which reduced the financial burden associated with recruitment.

Compounded rapamycin formulations have been reported to exhibit substantially lower bioavailability relative to generics, with some estimates suggesting approximately threefold lower systemic exposure [70]. Accordingly, a weekly dose of 20 mg of compounded rapamycin is estimated to achieve systemic exposure comparable to approximately 6 mg/week of generic rapamycin [41]. Compounded formulations can be more affordable and accessible; however, sourcing derived from licensed providers with complete purity data is essential. Clinicians should recognize the bioavailability differences between the generic and compounded formulations to achieve correct dosing, maximize clinical benefit, and reduce adverse events [71].

The sex-specific dosing regimens used in this study (20 mg/week for males and 15 mg/week for females) were informed by published findings from the PEARL ageing study (NCT04488601), conducted in healthy adult populations; however, the biological basis for sex-specific dosing remains speculative [41, 46]. In clinical practice, individualized dosing strategies should incorporate patient age, cardiac history, standard hematology and metabolic laboratory values, longitudinal whole-blood sirolimus peak/trough concentrations, CYP3A4/P450 genotype status, dietary influences related to drug

administration, and all potential drug-drug interactions from concomitant medications [72]. Importantly, abrupt discontinuation of low-dose rapamycin is not contraindicated, and although dose-escalation strategies are generally recommended to improve tolerability, a prolonged ramping period may not be necessary in all patients. Comprehensive information regarding known drug interactions is available through the U.S. Food and Drug Administration prescribing guidance and should be considered alongside medical history and routine safety laboratory assessments [70, 73, 74].

Longitudinal safety monitoring revealed no clinically significant alterations in hemoglobin A1C, insulin, complete blood count, platelet count, lipid profile, or hs-CRP throughout the study period. Similarly, comprehensive metabolic analyses demonstrated no significant changes in lipid, cholesterol, glucose, or inflammatory biomarker profiles during the 90-day treatment period. While it has been widely reported that chronic rapamycin therapy can result in hyperlipidemia, impaired glucose metabolism, and immunosuppression [75], we did not observe clinically significant changes in these safety laboratory parameters, nor did we observe an increase in common or opportunistic infections within our study population. Although our sample size remains relatively small, it is conceivable that the low-dose, intermittent administration schedule utilized in this study may mitigate some of the adverse metabolic and immunosuppressive effects that have been reported in transplant populations and animal models exposed to chronic high-dose and/or higher frequency.

Our pharmacokinetic analyses in this study, which utilized non-fasting peak and trough whole-blood sirolimus measurements, confirmed detectable systemic drug exposure and were consistent with the established pharmacokinetic profile of sirolimus, including an approximate peak concentration (C_{max}) at 2 h post-dose, an elimination half-life of approximately 66 h, and primary metabolism through CYP3A4/P450-associated pathways.

Whole blood sirolimus concentrations taken at both “peak” (2 h after dose administration) and “trough” (taken approximately 48 h after dose administration) demonstrated substantial inter-individual variability across the study population as reported by others. Sirolimus results are described in the Results section. Notably, neither peak nor trough sirolimus levels correlated with clinical response categories (responders, partial responders, or non-responders) [41]. Due to significant intra- and inter-individual variability in sirolimus pharmacokinetics, drug levels often correlate poorly with clinical response, even in transplant patients, and thus require regular drug therapeutic monitoring. As such, it is not surprising that trough levels did not correlate with symptom improvement in our low-dose, once-weekly dosing paradigm [74, 76].

Future trials investigating the pharmacokinetics of low-dose rapamycin should carefully account for CYP3A4/P450 genetic status, fasting or the type of dietary intake at the time of dosing, and the timing of blood collection relative to drug administration. Sirolimus has low average bioavailability ranging between 14 and 20% due to its large and highly lipophilic structure. Oral formulations face additional challenges due to high CYP3A4/P450 expression in the small intestine and liver. Studies have shown that administration of oral sirolimus with a high fat meal can slow absorption and reduce the peak C_{max} , and the ratio of C_{max} to AUC lengthening the time to achieve max concentration and can moderately increase the bioavailability up to 35% compared to when administered under fasting conditions [77]. In addition to better drug absorption and increased bioavailability, we found anecdotal evidence in our cohort that a high fat meal at the time of dose appeared to reduce GI symptoms (gas, diarrhea and bloating).

Clinically, treatment with compounded rapamycin produced broad improvements across multiple validated measures of ME/CFS symptom severity, physical function, cognition, and quality of life. Significant improvements were observed in physical fatigue-related domains measured by BAS, aggregate MFI scores, SSS fatigue, PEM, orthostatic intolerance (OI), disturbed sleep, and several SF-36 physical health domains, including bodily pain and physical functioning. Improvements in emotional and cognitive domains including mental fatigue, communication, social functioning, and concentration were detectable as early as T1 and continued through T3. Notably, additional non-canonical fatigue assessment instruments, including FACT-G and FACIT-F, also demonstrated significant improvements in physical, emotional, social, and functional well-being, suggesting that rapamycin treatment broadly improved fatigue-associated symptom burden and overall quality of life in ME/CFS subjects.

Wearable device data were collected from a subset of participants; however, the absence of a standardized wearable platform limited meaningful cross-subject comparisons. This limitation was driven in part by budgetary constraints, but reflects broader challenges associated with wearable implementation in ME/CFS research. Current commercially available devices vary substantially in cost, battery requirements, software architecture, and accessibility to raw physiologic data, limiting inter-device compatibility for advanced metrics such as heart rate variability, sleep architecture, and estimated VO2 max. In addition, few platforms have been validated specifically for long-term monitoring in ME/CFS or other highly disabled patient populations and are designed to track and promote physical exercise which is contraindicated in ME/CFS [78]. Future studies will require the implementation of a single, validated wearable platform suitable for decentralized clinical monitoring and compatible with the functional limitations of this patient population. Until such approaches become standardized, validated self-reported outcome instruments, including those used in the present study, remain the primary measures of clinical efficacy in ME/CFS research.

Taken together, the study enhancements implemented in Phase II resulted in a substantially higher study completion rate (83%) compared to Phase I (47%) and we found that compounded rapamycin was safe and generally well tolerated among a representative cohort of ME/CFS patients. While these results show promise, the lack of a placebo control in this trial must again be acknowledged.

Viral versus non-viral cohorts

Acute viral illnesses, including SARS-CoV-2 infection, are a well-recognized precipitating factor for a subset of ME/CFS cases. Accordingly, we investigated whether participants reporting a viral onset of illness demonstrated a differential response to rapamycin therapy compared to those reporting a non-viral onset. Among trial completers and active participants ($N=76$), individuals with a reported viral onset history ($N=39$) demonstrated a higher proportion of clinical responders to rapamycin therapy relative to the non-viral onset cohort. Specifically, 48.72% of participants in the viral-onset group were classified as responders, whereas only 18.92% of participants in the non-viral onset group achieved responder status. In contrast, partial responder classifications were more frequently observed in the non-viral onset cohort. Although these subgroup analyses remain exploratory and are limited by sample size and self-reported illness onset histories, the findings raise the possibility that viral-associated ME/CFS endophenotypes (including Long-COVID patients who meet ME/CFS criteria)

may demonstrate increased therapeutic outcomes using mTOR inhibitors.

Evaluation of functional outcome of phase-II rapamycin trial

While previous results indicated that rapamycin may boost autophagy in ME/CFS patients by increasing BCLN-1 and suppressing S258PATG13, molecular changes do not always equate to functional cell recovery. Therefore, we conducted functional assays to assess rapamycin's impact on metabolic recovery.

First, we examined mitochondrial energy metabolism in PBMCs from ME/CFS subjects

Seahorse OCR analysis showed rapamycin significantly enhanced mitochondrial reserve capacity, suggesting improved energy metabolism under stress. Additionally, higher reserve capacity negatively correlated with PEM, OI, and sleep disturbances.

Second, we evaluated rapamycin's effect on inflammation in myeloid cells

ME/CFS is linked to impaired oxygen utilization, contributing to OI and PEM. In vitro, exposing myeloid cells to ME/CFS serum under hypoxia increased inflammatory markers (CD40, iNOS). Similarly, baseline plasma upregulated M1 antigen expression and iNOS. However, plasma from rapamycin-treated patients reduced M1 antigens and iNOS while elevating anti-inflammatory M2 marker CD206, highlighting rapamycin's potential to mitigate inflammation in ME/CFS.

Third, we investigated the impact of rapamycin on purine metabolism and its relationship with inflammation in ME/CFS

Recent studies indicate that dysregulation of purine metabolism may contribute to the pathogenesis of ME/CFS. While the underlying mechanism remains unclear, our previous research [35] demonstrated that the impaired aerobic glycolysis followed by enhanced pentose phosphate pathway activity may increase the biosynthesis of purines associated with ME/CFS pathogenesis. In this process, IMP, a key precursor in purine biosynthesis, is enzymatically converted to XMP by IMPDH2. Our previous findings revealed that myeloid cells exhibit elevated IMPDH2 expression under hypoxic conditions, resulting in increased conversion of IMP to XMP.

Notably, our LCMS assessment of purine metabolites, complemented by flow cytometry analysis of IMPDH2 activity, showed that rapamycin treatment significantly reduced the activation of IMPDH2 in ME/CFS subjects. Although the underlying mechanism remains unclear, recent research indicates that inhibition of mTOR may contribute to the activation of the cellular phosphatase

PP2A [79]. PP2A has been demonstrated to dephosphorylate IMPDH2 at Ser122, resulting in its inactivation [55]. The diminished activity of IMPDH2 was further confirmed by the reduced conversion of IMP (substrate) to XMP (product) as quantified by LCMS study in T3 ME/CFS subjects compared to baseline control (BSL). Other studies [45, 80, 81] suggest that mTOR inhibition can suppress tetrahydrobiopterin (BH4) biosynthesis. Our results suggest that the elevated BH4 biosynthesis identified in our cohort of ME/CFS subjects is likely attributed to the increased IMPDH2 activity resulting from mTOR pathway activation. The relevance of elevated IMP level by rapamycin treatment may be attributed to the increased ATP biosynthesis via enzymic reaction of adenylosuccinate synthetase (ADSS) [82, 83]. ADSS protein is strongly expressed in muscle tissue and in fact, the genetic suppression of ADSS variant 1 (ADSS1), was found to be responsible for myopathy and neuromuscular abnormalities [84].

Furthermore, we found that rapamycin treatment decreased the bioavailability of HPX, another metabolic end-product of IMP. Since HPX upregulation promotes peroxide radical formation and oxidative stress, the reduction of HPX in T3 samples may indicate that the rapamycin-mediated suppression of HPX is critical in mitigating oxidative stress in ME/CFS subjects.

Fourth, we examined whether disrupted purine metabolism in ME/CFS subjects impacts the glycolytic index in myeloid cells

Inflammatory cells require substantial ATP, relying on glycolysis and mitochondrial oxidative phosphorylation for energy. During inflammation, oxidative stress impairs mitochondria, causing cells to shift toward glycolysis—a trend seen in M1 macrophages. However, this metabolic adaptation hasn't been confirmed in ME/CFS pathogenesis.

Our study assessed how plasma purines from ME/CFS subjects affect microglial glycolysis. ECAR analysis after mitochondrial disruption under hypoxia showed that BSL plasma strongly reduced glycolytic capacity and reserve, indicating suppressed maximal glycolysis. In contrast, T3 plasma restored glycolytic efficiency, suggesting rapamycin may improve glycolysis under stress.

Silencing P_2X_7 with siRNAs enhanced cells' response to BSL plasma, nearly restoring glycolytic capacity, though buffer capacity for acute energy surges remained limited. Conversely, P_2X_4 inhibition further decreased basal glycolysis without improving stress-related glycolytic parameters. These findings imply that elevated purines in ME/CFS may suppress glycolysis via P_2X_7 signaling, but this pathway may not operate in microglia exposed to plasma from rapamycin-treated subjects.

Fifth, we also studied purinergic effects on mitochondrial respiration in hypoxic microglia

While mitochondrial respiration has been explored in ME/CFS, its regulation by purinergic signaling is unclear. We found BSL plasma significantly reduces mitochondrial respiration compared to HC plasma, and this suppression persists after inhibiting P_2X_4 or P_2X_7 . Inhibiting P_2X_4 restored mitochondrial OXPHOS during stress, but not basal respiration. Rapamycin-treated plasma improved mitochondrial respiration in hypoxic microglia, likely independent of P_2X_7 signaling.

Study limitations and future directions

Limitations of the current study

Several limitations should be acknowledged in the present investigation. *First*, the study was not designed as a placebo-controlled trial. Instead, each participant served as their own control, with baseline (BSL) measurements compared to those obtained after treatment at the T3 stage. This approach may introduce certain biases and limit the ability to fully account for placebo effects or other confounding variables. *Second*, while the study provides evidence of dysregulated purine metabolism in ME/CFS subjects, the specific purine intermediates responsible for interacting with P_2X_4 and P_2X_7 receptors to drive oxidative stress and inflammation have not been identified. As such, a more detailed mechanistic investigation is necessary to characterize additional purine intermediates and their regulatory roles in this context. *Third*, although P_2X_4 and P_2X_7 receptors are two of the most extensively studied inflammatory purinergic receptors, other purinergic receptors, such as members of the P_2Y family, were not examined within the current scope of this research. Additional studies are needed to explore the potential involvement of these receptors in ME/CFS pathogenesis. *Fourth*, in some cases, results were obtained from responders, which introduces selection bias into the study. *Fifth*, the multi-modal responses to rapamycin—including enhancement of mitochondrial energy metabolism, suppression of inflammatory phenotypes, and alterations in purinergic signaling—may be associative effects rather than causal relationships. Further studies and in-depth molecular analyses will identify more causal mechanisms. *Finally*, no prior power calculations for the LCMS, FACS, or Seahorse OCR assays were done. These pilot experiments were designed as mechanistic, hypothesis-driven analyses, and the sample size of eight ($N=8$) was accepted for the methodological rigor of our current study.

Future Directions

As a future direction for this project, we plan to recruit participants for a double blinded placebo-controlled step-wedge clinical trial to strengthen the validity of our

findings. The step wedge design aims to ease enrollment by ensuring that all participants are guaranteed access to the study drug and not just the placebo; an important factor when studying FDA-approved agents that are widely available. Ongoing studies are also focused on assessing the effects of compounded rapamycin on the regulation of autophagy markers, including Beclin-1 and S258P ATG13, to further elucidate the therapeutic potential and molecular mechanisms of rapamycin in ME/CFS. Rapamycin modulates immune responses by suppressing B cell activation [85], facilitates $CD4^{+ve}$ cell expansion via activation of regulatory T cells [86], and promotes Th1/Tc cell activation [87], which may collectively result in reduced inflammation, immunomodulation, and immune senescence that is beneficial not only for the suppression of autoimmune response [88], but also enhanced immune function during anti-tumor immunity [89]. To investigate T cell regulation and metabolism in ME/CFS after rapamycin treatment, a high-resolution transcriptomics study will be conducted.

Collectively, our current study demonstrates evidence that plasma-derived purines in ME/CFS subjects may mediate purinergic signaling through the $P_2X_{4/7}$ receptor pathway. This signaling appears to reduce mitochondrial oxidative phosphorylation under stress, which in turn promotes the development of an inflammatory phenotype in myeloid cells.

Abbreviation

ME/CFS	Myalgic Encephalomyelitis/Chronic Fatigue Syndrome
PEM	Post-exertional malaise
OI	Orthostatic Intolerance
mTOR	the mammalian target of Rapamycin
IMP	Inosine monophosphate
XMP	Xanthosine 5 monophosphate
HPX	Hypoxanthine
IMPDH2	IMP dehydrogenase 2
MFI	Multidimensional fatigue index
BAS	Bell activity scale
SSS	Symptom Severity Scale
SF-36	Short Form 36-Item Health Survey
$P_2X_{4/7}$	P_2X purinoceptor 4 or 7
OXPHOS	Oxidative phosphorylation
HC	Healthy control
BSL	baseline ME/CFS subjects
T1/T2/T3	30/60/90 days rapamycin-treated ME/CFS subjects

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-026-08575-3>.

Supplementary Material 1

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and sentence structure when drawing scientific conclusions, the copilot AI application was occasionally employed and approved after careful consideration.

Author contributions

A.R. and C.G.G. conceived the study. B.G., S.B., M.A.T., B.T.R., and S.M. performed laboratory research. Clinicians, including B.C., L.B., B.Y., J.B., D.P., S.G. and D.K. recruited patients, prescribed drugs, and designed the research. A.R., B.G., C.G.G., B.T.R., and S.B. curated the data and wrote the manuscript. B.G., C.G.G., and A.R. supervised and facilitated collaborative research between UWM and Simmaron, regulated mobile phlebotomy services, acquisition of biospecimens, and development of biobanks at the University of Wisconsin Milwaukee. All the authors have read and agreed to the published version of the manuscript.

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Data availability

The revised, updated and full version of data was shared in Mendeley online data repository and available via the link with doi: 10.17632/tkgms4pmdz.3

Declarations

Ethics and institutional review board

The study was conducted under a central protocol (SRI-RP-2023-1) and all sites were approved by the Western Institutional Review Board (WIRB Study Number: 1361805). The study complies with the standard ethical principles developed and maintained by World Medical Association (WMA) under the declaration of Helsinki. The IBC protocol (Protocol # B26.015) for conducting basic research on PBMCs has been approved by University of Wisconsin-Milwaukee. Simmaron Research Inc. is the study sponsor, and the trial was registered on ClinicalTrials.gov (NCT06257420). No part of this research article, including images and texts, has been published or reproduced in any book, article, or other published literature. The images were original and not copied from any other source.

Consent to participate in the trial

All participants consented to the study.

Informed consent

All authors provided consent to publish this original research article.

Competing interests

The authors declare no conflicts of interest.

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